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Clustering of architectural floor plans: A comparison of shape representations

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ABSTRACT

Generative design methods are able to produce a large number of potential solutions of architectural floor plans, which may be overwhelming for the decision-maker to cope with. Therefore, it is important to develop tools which organise the generated data in a meaningful manner. In this study, a comparative analysis of four architectural shape representations for the task of unsupervised clustering is presented. Three of the four shape representations are the Point Distance, Turning Function, and Grid-Based model approaches, which are based on known descriptors. The fourth proposed representation, Tangent Distance, calculates the distances of the contour's tangents to the shape's geometric centre. A hierarchical agglomerative clustering algorithm is used to cluster a synthetic dataset of 72 floor plans. When compared to a reference clustering, despite good perceptual results with the use of the Point Distance and Turning Function representations, the Tangent Distance descriptor (Rand index of 0.873) provides the best results. The Grid-Based descriptor presents the worst results.

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1. Introduction

Generative design methods are commonly used in architectural design. These methods have several applications in the design of structural elements, facade layout, space planning, optimisation of building form, replication of architectural styles, and urban design. The main goal is to assist building design practitioners in exploring a larger set of solutions, which a traditional trial-and-error process could never achieve. However, one of the drawbacks is that they may produce an excessive number of solutions for a human to cope with; moreover, it is just not feasible to rate solutions according to a performance criterion and then select the top-ranked ones, especially for unclear and subjective problems. An alternative approach is to organise the generated data into groups determined by common features. This allows the decision-maker to compare group types before analysing specific solutions. Therefore, to facilitate the decision-maker's task of comparison and selection, this paper presents an unsupervised clustering technique using four different shape representations. The method and the performance of these shape descriptors is analysed in a computer generated architectural floor plan showcase.

This is a typical task for machine learning techniques. In the field of machine learning there are two main subfields dealing with organisation of data: classification and clustering. While the former is used to label data according to pre-defined classes, the latter deals with unlabelled data and the task is usually to create partitions in the data while making coherent groups according to some defined metric. This is a process of identifying structures in unlabelled datasets regardless of the data type. Han and Kamber [1] classified clustering techniques into five categories: partitioning methods, hierarchical methods, density-based methods, grid-based methods, and model-based methods.

Clustering techniques have been applied in diverse areas. Some of the most relevant applications include the classification of textual documents [2], document navigation for search engine optimisation [3–5], resource project scheduling [6], point cloud simplification [7,8], time series analysis and clustering [9], image clustering [10], face expression [11], database retrieval of mechanical objects [12,13], and sketch recognition [14].

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建筑平面图的聚类：形状表示方法的比较

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摘 要

生成式设计方法能够生成大量潜在的建筑平面方案，这可能会使决策者难以应对。因此，开发能够以有意义的方式组织生成数据的工具至关重要。本研究针对无监督聚类任务，对四种建筑形态表征方法进行了比较分析。其中三种表征方法分别为基于已有描述子的点距离法、转向函数法和基于网格的模型法；第四种提出的切线距离法则计算轮廓切线到形体几何中心的距离。我们采用层次凝聚聚类算法对包含72个平面方案的合成数据集进行聚类。与参考聚类结果相比，尽管点距离法和转向函数法在视觉感知上表现良好，但切线距离描述子（Rand指数为0.873）仍取得了最佳效果，而基于网格的描述子则表现最差。

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1. 引言

生成式设计方法在建筑设计中得到广泛应用。这些方法在结构构件设计、立面布局、空间规划、建筑形态优化、建筑风格再现以及城市设计等领域具有多种应用。其主要目标是帮助建筑设计从业者探索更为广泛的方案空间，而这一目标是传统的试错法所难以实现的。然而，其一个局限在于，生成的方案数量可能过多，超出人工处理的负荷；此外，针对模糊且主观性强的问题，按照性能指标对所有方案进行评分并从中遴选最优方案也并不现实。一种替代方案是根据共同特征将生成的数据划分为若干组，从而使决策者能够在分析具体方案之前先对各组类型进行比较。因此，为便于决策者开展比较与选择工作，本文提出了一种基于四种不同形状表示的无监督聚类技术。该方法及其所采用的形状描述子的性能在一组计算机生成的建筑平面图示例中得到了分析。

这是一项典型的机器学习任务。在机器学习领域，有两个主要的子领域涉及数据的组织：分类与聚类。前者用于根据预先定义的类别对数据进行标注，而后者则处理未标记的数据，其任务通常是在数据中划分出若干子集，并依据某种给定的度量标准形成具有内在一致性的簇。这一过程旨在识别无标签数据集中的结构，且不依赖于数据类型。Han和Kamber[1]将聚类技术划分为五大类：划分方法、层次方法、基于密度的方法、基于网格的方法以及基于模型的方法。

聚类技术已被广泛应用于多个领域。其中一些最为重要的应用包括：文本文档的分类[2]、用于搜索引擎优化的文档导航[3–5]、资源项目调度[6]、点云简化[7,8]、时间序列分析与聚类[9]、图像聚类[10]、人脸表情识别[11]、机械对象的数据库检索[12,13]，以及草图识别[14]。

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The clustering of objects, according to their shape, has also been previously applied in diverse fields. The correct representation of the shape has a significant impact on the matching correctness of the algorithms [15]. For instance, Chang et al. [16] proposed a shape recognition scheme where the representation corresponds to the distance of feature points in the shape's boundary to the centroid. This shape representation presents the property of being invariant to translation as the boundary is fixed in relation to the centroid independently of its global position. As the distances of the feature points are ordered and divided by a minimum distance, this also results in invariance to scaling, rotation, and reflection. Instead of only considering the shape feature points, Yankov and Keogh [17] used the entire contour for the shape representation and a nonlinear reduction technique to cluster pathological cells.

Arkin et al. [18] represented a polygonal shape by its turning function. The shape descriptor consists in measuring the angle of the counter-clockwise tangent to the x-axis in each of the feature points in the polygon. Therefore, the values vary between $-\pi$ and π. As the polygon is scaled to have a length of 1, in addition to being translation invariant, the representation is also invariant to scaling. However, results depend on the starting point and the polygon's rotation and reflection.

Sajjanhar and Lu [19] suggested a grid-based representation where a shape is placed, rotated, and scaled to fit a square grid. For each cell in the grid a binary value is determined: 0 for empty and 1 for filled. Although this representation guarantees translation and scale invariance, if the grid is adaptive, the scaling is only invariant to one of the axes—the rotation invariance is dependent on the rotation of the grid to match the same shape orientation. Also, as may be expected, the results vary according to the grid size, as this changes the capability to capture the shape's details.

Siddiqi et al. [20] used a shock graph to capture the effects on the bounding contours of the singularities in the shape structure. The graph is determined according to a set of rules in a shock graph grammar which reduces it to a rooted shock tree. A recursive algorithm is then used to match two shock trees, starting from the root and proceeding through the subtrees in a depth-first approach.

Belongie et al. [21] presented an approach to measure similarity of shapes by considering the distribution of the remaining points in each reference point. As corresponding points in two similar figures have similar contexts, a transformation is used to align two shapes. The dissimilarity between them is calculated by summation over the errors between the corresponding points in the transformation.

Aiming to retrieve shapes from a database, which are similar to a query shape, Tan et al. [22] proposed a new representation based on a centroid-radii approach. According to the authors, this approach allows the modelling of convex, concave, and hollow shapes. The representation consists of a set of vectors, each one measured at regular intervals from the centroid of a concentric ring.

In Klassen et al. [23], the shapes are considered to be planar closed curves represented either as direction functions or as curvature functions. In this manner, shapes may be modelled as stretchable, compressible, and bendable strings along their extensions that are constructed from spaces of parametric curves [24,25]. Geodesics are used to determine the dissimilitude between shapes.

Ling and Jacobs [26] classified shapes by using an inner-distance to build the shape representation of the structure or articulation parts. The inner-distance is the length of the shortest path between two reference points on the shape boundary and allows the creation of articulation invariant representations.

Shen et al. [27] proposed a method to group planar figures by their skeleton graph. The clustering is carried out by determining the common internal shape structure that belongs to the same cluster. The data is grouped by using an agglomerative clustering algorithm.

In architecture, Cha and Gero [28] investigated shape patterns to determine if any similarities, relationships, and physical properties could be recognised. de las Heras et al. [29] used run length histograms as a perceptual representation of floor plans made by architects. This approach allows the retrieval of designs with similar properties from a database. Dutta et al. [30] used a graph-based method to identify symbols in floor plans such as furniture and openings.

However, despite all of the mentioned approaches/methods, the use of clustering techniques has yet to be used to group designs in the case of automatic generation of floor plans. In a previous study, Sousa-Rodrigues et al. [31,32] conducted an online survey directed at design and construction experts—mostly architects, engineers and architecture undergraduates—in which the majority of respondents considered the overall shape of floor plans as the most important similitude feature. This highlights the importance of having perceptually accurate algorithms for the automation of this task.

In this paper, four shape representations are studied as floor plan design descriptors under the same settings. All descriptors are vectors of similar length, and all are used to partition the same dataset with the same clustering algorithm. Three of the four shape representations are known descriptors: these are the distance to centroid [16], the Turning Function [18], and the Grid-Based model [19]. The fourth and last shape descriptor is a novel representation specifically created to capture orthogonal floor plan shapes. It consists in calculating the distance of the tangent lines to the geometric centre of the shape. The clustering procedure is an agglomerative hierarchical algorithm with Ward linkage [33] and Euclidean distance as a dissimilarity measure. The advantages and disadvantages of each shape representation are analysed in a showcase with 72 floor plan designs. These designs were generated using a specific algorithm, named Evolutionary Program for the Space Allocation Problem (EPSAP) [34–36]. The EPSAP algorithm generates alternative floor plans according to the user's specifications.

After this introductory section, Section 2 describes the methods applied to the clustering of the floor plans designs. In Section 3 the results for a showcase of a single-family house are presented and compared to a reference clustering partition. The discussion of the relevant results follows in Section 4, as well as the analysis of the applicability of the descriptors. Finally, conclusions are drawn and future work is outlined in Section 5.

2. Methodology

To determine the most suitable shape representation to be used in the cluster of orthogonal floor plans, three shape descriptors inspired by previous works and one new descriptor were implemented. These descriptors have the same vector length and shape matching algorithm using the Euclidean distance to calculate the dissimilitude between the shapes. Therefore, the computational burden is equal for the four approaches. A specific algorithm generated a dataset of floor plan designs. This synthetic dataset does not require a pre-processing mechanism for denoising the shapes, nor the application of a dimensionality reduction technique. Therefore, the focus is on the perceptual quality of the results of each shape descriptor.

2.1. Shape representation

The representation of continuous features plays an important role in machine learning techniques, either because the machine learning technique itself requires a nominal feature space—nominal features describe qualitative aspects that do not share a natural ordering relationship—or because discretisation allows for better results in the machine learning technique. The research on dataset discretisation for machine learning is vast and beyond the scope of this paper, but it is important to mention that such algorithms usually aim to maximise the interdependency between discrete attribute values and class labels, as this minimises the information loss due to

根据物体的形状对其进行聚类的方法也已在多个领域得到应用。形状的正确表示对算法匹配的准确性具有重要影响[15]。例如，Chang 等人[16]提出了一种形状识别方案，其中的形状表示方式为：将形状边界上各特征点到形心的距离作为表征。由于边界相对于形心的位置是固定的，且不受其全局位置的影响，这种形状表示具有平移不变性。此外，通过对特征点距离进行排序并除以最小距离，该表示还具备尺度、旋转及反射不变性。与仅考虑形状特征点不同，Yankov 和 Keogh[17]则采用整个轮廓作为形状的表征，并结合非线性降维技术对病理细胞进行聚类。

Arkin 等人[18]通过转向函数来表征多边形的形状。该形状描述子通过测量多边形各特征点处切线相对于 v1 轴的逆时针方向角度来构建，因此其取值范围在 v2 和 v3 之间。当将多边形缩放至长度为 1 时，该表示不仅具有平移不变性，还具有尺度不变性。然而，所得结果仍依赖于起始点以及多边形的旋转和镜像变换。

Sajjanhar 和 Lu [19] 提出了一种基于网格的表示方法，其中将形状放置、旋转并缩放以适配一个方形网格。对于网格中的每个单元格，均赋予一个二值：空单元格为0，填充单元格为1。尽管这种表示方法能够保证平移不变性和尺度不变性，但如果采用自适应网格，则尺度不变性仅对其中一个坐标轴成立——而旋转不变性则依赖于网格的旋转以匹配同一形状的朝向。此外，正如所预期的那样，结果会随网格尺寸的变化而有所不同，因为这会影响对形状细节的刻画能力。

Siddiqi 等人[20]采用冲击图来刻画奇异点在形状结构边界轮廓上的影响。该图依据冲击图文法规则集确定，并可归约为一棵带根的冲击树。随后，采用递归算法对两棵冲击树进行匹配，自根节点开始，以深度优先的方式逐层遍历各子树。

Belongie 等人[21]提出了一种通过考察每个参考点处剩余点的分布来度量形状相似性的方法。由于两个相似图形中的对应点具有相似的局部结构，因此采用一种变换来对齐这两个形状，并通过累加变换过程中对应点之间的误差来计算它们之间的不相似度。

为了从数据库中检索与查询形状相似的形状，Tan 等人[22]提出了一种基于质心-半径方法的新表示。根据作者的论述，该方法能够对凸形、凹形及具有空腔的形状进行建模。该表示由一组向量构成，每个向量均以等间隔方式从同心环的质心处测量得到。

在Klassen等人的研究中[23]，形状被视为由方向函数或曲率函数表示的平面闭合曲线。据此，形状可被建模为沿其延伸方向具有可拉伸、可压缩及可弯曲特性的弹性曲线，这些曲线由参数化曲线空间构成[24,25]。测地线被用于度量形状之间的差异性。

Ling和Jacobs[26]通过引入内距离来构建结构或关节部件的形状表示，从而对形状进行分类。内距离是指形状边界上两个参考点之间最短路径的长度，这一度量能够用于构建对关节变化具有不变性的形状表示。

Shen 等人[27]提出了一种基于骨架图对平面图形进行分组的方法。该方法通过识别属于同一簇的共有的内部形状结构来进行聚类，并采用凝聚式聚类算法对数据进行分组。

在建筑学领域，Cha 和 Gero [28] 研究了形态模式，以确定是否存在任何相似性、关联性以及物理特性。可以被识别。de las Heras 等人[29]将游程直方图用作建筑师所绘制的平面图的感知表示，该方法允许从数据库中检索具有相似特征的设计。Dutta 等人[30]则采用基于图的方法来识别平面图中的家具、洞口等符号。

然而，尽管已提出了上述各种方法与途径，聚类技术仍未被用于在平面图自动生成过程中对设计方案进行分组。在先前的一项研究中，Sousa-Rodrigues 等人[31,32]针对设计与施工领域的专家——主要包括建筑师、工程师及建筑学本科生——开展了一项在线调查，结果显示，大多数受访者认为平面图的整体形状是最为重要的相似性特征。这一发现凸显了开发感知上准确的算法以实现该任务自动化的必要性。

在本文中，我们在相同实验条件下研究了四种用于平面布局设计的形状表征方法。所有表征均为长度相近的向量，并均采用相同的聚类算法对同一数据集进行划分。这四种形状表征中有三种为已知方法，分别为：到形心的距离[16]、转向函数[18]以及基于网格的模型[19]。第四种也是最后一种形状表征是一种新提出的表示方法，专门用于刻画正交型平面布局的几何特征；其具体做法是计算各切线到形状几何中心的距离。聚类过程采用基于Ward链接法的凝聚式层次聚类算法[33]，并以欧氏距离作为相异度度量。我们通过包含72个平面布局方案的案例展示了每种形状表征的优缺点。这些方案均由一种名为“空间分配问题演化规划”（EPSAP）[34–36]的特定算法生成，该算法可根据用户的具体需求生成多种备选平面布局方案。

在本导论部分之后，第2节介绍了用于对平面布局设计方案进行聚类的方法。第3节展示了针对一栋独户住宅的示例结果，并将其与参考聚类划分进行了比较。第4节对相关结果进行了讨论，并分析了各描述符的适用性。最后，第5节总结了研究结论，并展望了未来的研究方向。

2. 方法论

为了确定在正交平面图聚类中应采用的最适形状表示方法，我们实现了三种受既有研究启发的形状描述子以及一种新的描述子。这些描述子具有相同的向量长度，并采用基于欧氏距离的形状匹配算法来计算形状之间的不相似度。因此，四种方法的计算开销相当。一个专门的算法生成了一个平面图设计数据集。该合成数据集无需进行形状去噪的预处理，也无需应用降维技术。因此，研究重点在于各形状描述子所得结果的感知质量。

2.1. Shape representation

连续特征的表示在机器学习技术中起着重要作用，原因在于：一方面，某些机器学习算法本身要求使用名义特征空间——名义特征用于描述不具备自然序关系的定性属性；另一方面，离散化处理往往能够提升机器学习算法的性能。关于面向机器学习的数据集离散化研究已十分丰富，超出了本文的讨论范围，但有必要指出的是，此类算法通常旨在最大化离散属性值与类别标签之间的互依性，因为这有助于最小化因

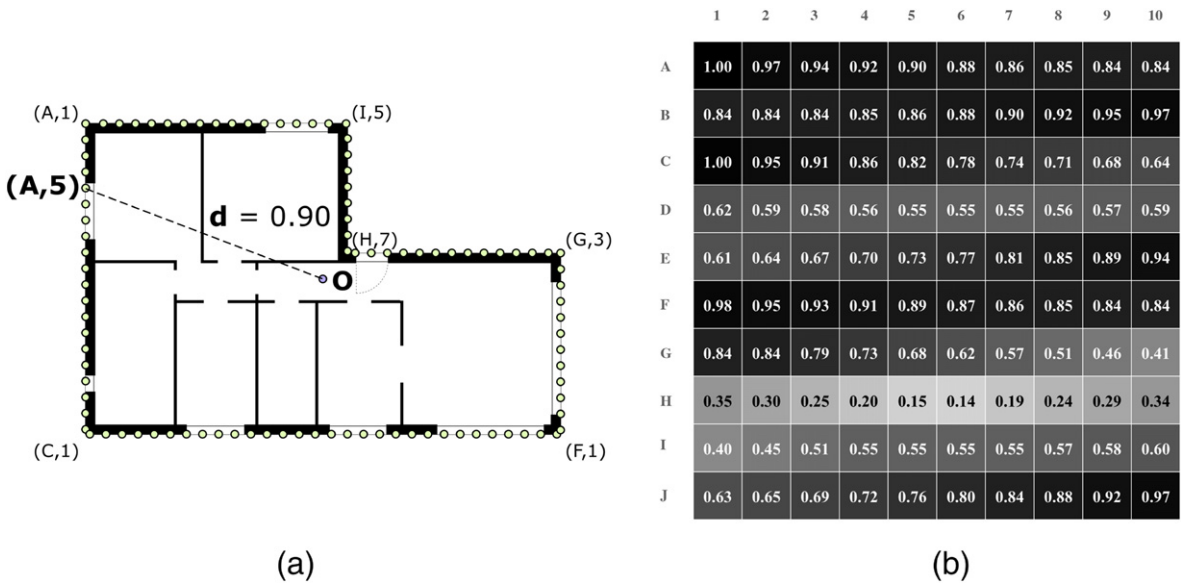


Fig. 1. Point Distance (PD) descriptor. (a) Example of the normalised distance for the point (A,5) with value of 0.90, which corresponds to its real distance divided by the longest distance of all silhouette points. The wall corners are marked with the matrix index to depict the counter-clockwise order of the feature points. (b) Vector in the form of a gradient matrix (white is 0 and black is 1) of the normalised distances.

the discretisation process. The process has to balance the trade-offs between these two goals and many studies have shown that several machine-learning techniques benefit from it [37–40].

In this study, the four descriptors are designed to have similar features. These are invariant to translation and scaling but sensitive to rotation and reflection. A descriptor variant that considers independent scaling of x-and y-coordinates was also analysed. The reason for these features is that, despite floor plans being generated on a blank canvas, human experts continue to have a notion of north-south and east-west framework, thus a rotated or a reflected floor plan is considered as an alternative design. Buildings have a strong relation with their environment and their form depends on the surrounding buildings, landscape, solar orientation, and so on. However, because there are no visual references around each floor plan, translation does not affect the human perception of that shape. As a result, rotation and reflection were considered as features that influence the clustering result. Nevertheless, invariance to rotation and reflection could be easily achieved by ordering the descriptor vector or considering the distribution of these values.

2.1.1. Point Distance (PD) descriptor

Based on Chang et al.'s [16] shape representation, the Point Distance (PD) descriptor has points marked on the shape silhouette at equal segment lengths. The starting point is the nearest shape perimeter point in relation to the top-left corner of the shape bounding box and the points are distributed in a counter-clockwise direction. Our implementation differs from Chang et al.'s representation as the reference point is not the shape's centroid, which is defined as the average of the x- and y-coordinates of all perimeter points, but instead considers the geometric centre of the bounding box as the reference point. The shape descriptor is then a vector of normalised values—corresponding to the distance from the reference point to the ordered perimeter points divided by the longest point distance.

Fig. 1a illustrates an example of the marked perimeter point (A,5) and its normalised distance to the centre (0.90). The example represents the descriptor variant where the x-coordinate and y-coordinate scales are preserved. Fig. 1b depicts the representation vector of normalised values ranging from 0 (white) to 1 (black) in a gradient matrix form,[1] where the first vector point is (A,1) and concludes in point (J,10). In the floor plan image, the wall corners are marked with the corresponding matrix point to depict the counter-clockwise order of the marked points.

2.1.2. Turning Function (TF) descriptor

The second shape descriptor is based on Arkin et al.'s [18] turning function. This consists in determining the counter-clockwise angle to the x-axis of a tangent in each feature point along the shape contour. The feature points are marked at equal distances.

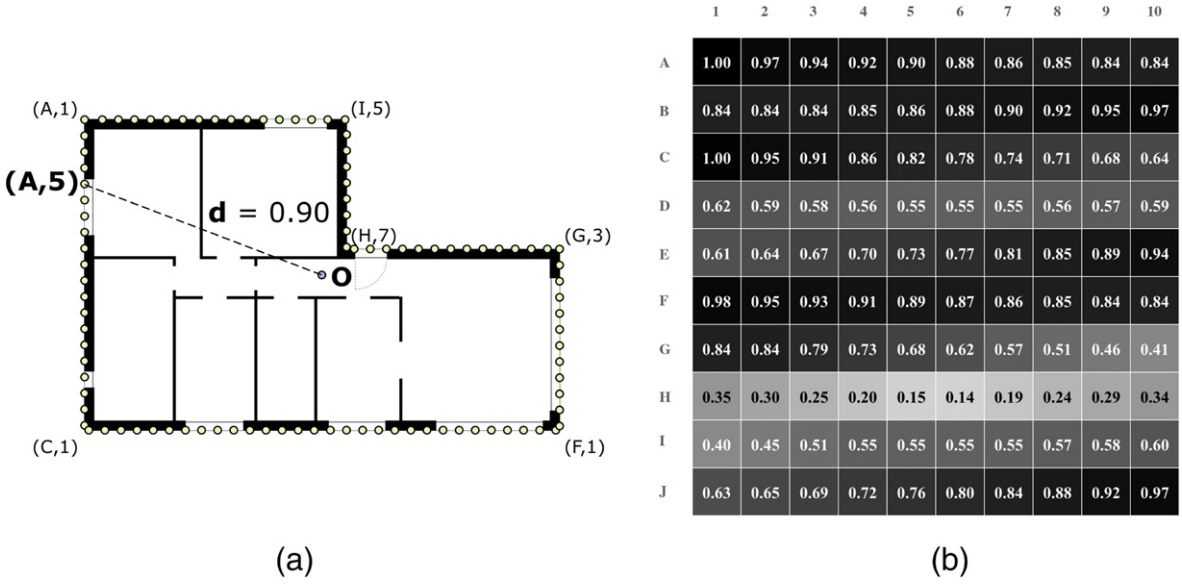
Fig. 2a depicts an example where the turning function angle is measured at point (B,3), with the value of $3\pi/2$, in the descriptor variant of preserved aspect ratio. The feature points start with the initial point (A,1), which is the nearest perimeter point to the top-left corner of the shape bounding box, and contours the shape silhouette in a counter-clockwise manner. Therefore, the values vary between 0 and 2π that are then normalised to have values ranging from 0 to 1. Fig. 2b illustrates the vector of the Turning Function (TF) descriptor as a gradient matrix. As the floor plans are orthogonal, the shape edges only take on four possible values $\{\pi/2, \pi, 3\pi/2, 2\pi\} = \{0.25, 0.50, 0.75, 1.00\}$.

2.1.3. Grid-Based (GB) descriptor

The Grid-Based descriptor is inspired on Sajjanhar and Lu's [19] work and consists in placing the shape under a square grid parallel to the exterior walls of the floor plans. For each cell in the grid, the centre may (1) or may not (0) be occupied by the shape area. The representation is a vector of binary values with the length equal to the number of cells. The values correspond to reading the grid from left-to-right and top-to-bottom.

Fig. 3a illustrates an example of a floor plan overlaid by a grid. In the example, point (B,8) has a value of 0 while (F,9) has a value of 1 depending on whether the floor plan area is under that cell centre or

[1] The gradient matrix of the four representations is used only for visual comparison of different floor plans. The agglomerative hierarchical algorithm uses each data point as a 1-dimensional vector.



	1	2	3	4	5	6	7	8	9	10
A	1.00	0.97	0.94	0.92	0.90	0.88	0.86	0.85	0.84	0.84
B	0.84	0.84	0.84	0.85	0.86	0.88	0.90	0.92	0.95	0.97
C	1.00	0.95	0.91	0.86	0.82	0.78	0.74	0.71	0.68	0.64
D	0.62	0.59	0.58	0.56	0.55	0.55	0.55	0.56	0.57	0.59
E	0.61	0.64	0.67	0.70	0.73	0.77	0.81	0.85	0.89	0.94
F	0.98	0.95	0.93	0.91	0.89	0.87	0.86	0.85	0.84	0.84
G	0.84	0.84	0.79	0.73	0.68	0.62	0.57	0.51	0.46	0.41
H	0.35	0.30	0.25	0.20	0.15	0.14	0.19	0.24	0.29	0.34
I	0.40	0.45	0.51	0.55	0.55	0.55	0.55	0.57	0.58	0.60
J	0.63	0.65	0.69	0.72	0.76	0.80	0.84	0.88	0.92	0.97

图 1. 点距离（PD）描述符。（a）点（A,5）的归一化距离示例，其值为0.90，等于该点的实际距离除以其长度。故障 所有轮廓点的距离。墙角以矩阵索引标记，用于表示特征点的逆时针排列顺序。（b）梯度形式的向量 伊恩特马化距离的二值矩阵（白色为0，黑色为1）。

离散化过程。该过程必须在上述两个目标之间进行权衡，且已有大量研究表明，多种机器学习技术可从中获益[37–40]。

在本研究中，所设计的四种描述子具有相似的特征：它们对平移和尺度变换具有不变性，但对旋转和反射较为敏感。此外，还分析了一种考虑x轴与y轴分别独立缩放的描述子变体。之所以采用这些特性，是因为尽管平面图是在空白画布上生成的，人类专家仍会基于南北和东西向的框架进行构想，因此旋转或反射后的平面图也被视为一种备选设计方案。建筑与其所处环境之间存在密切关联，其形态受周边建筑、地形地貌、日照朝向等因素的影响。然而，由于每张平面图周围缺乏视觉参照物，平移并不会改变人们对该形状的感知。因此，旋转与反射被视为会影响聚类结果的重要特征。不过，通过对描述子向量进行排序或考察这些值的分布，可以较为容易地实现对旋转与反射的不变性。

2.1.1. Point Distance (PD) descriptor

基于Chang等[16]提出的形状表示方法，点距离（PD）描述子在形状轮廓上按等分段长度标记若干点。起始点为形状包围盒左上角所对应的最近的形状周界点，各点沿逆时针方向分布。我们的实现与Chang等的表示方法有所不同：参考点并非形状的质心——即所有周界点x和y坐标均值——而是采用包围盒的几何中心作为参考点。由此得到的形状描述子为一个归一化向量，其元素值为从参考点到各有序周界点的距离除以最大点间距离所得的结果。

图1a示出了标记的周界点（A,5）及其相对于中心的归一化距离（0.90）的一个示例。该示例代表了在描述符变体中，x坐标与y坐标的比例得以保持的情形。图1b则以渐变方式展示了归一化值的表示向量，其取值范围从0（白色）到1（黑色）。

以矩阵形式表示，其中第一个向量点为(A,1)，终止于(J,10)。在平面图中，墙体的各角点均标注了相应的矩阵点，以示意所标记点的逆时针排列顺序。

2.1.2. Turning Function (TF) descriptor

第二种形状描述子基于Arkin等人[18]提出的转向函数。该方法通过确定沿形状轮廓上各特征点处的切线相对于x轴的逆时针角度来实现。这些特征点按等距分布进行标记。

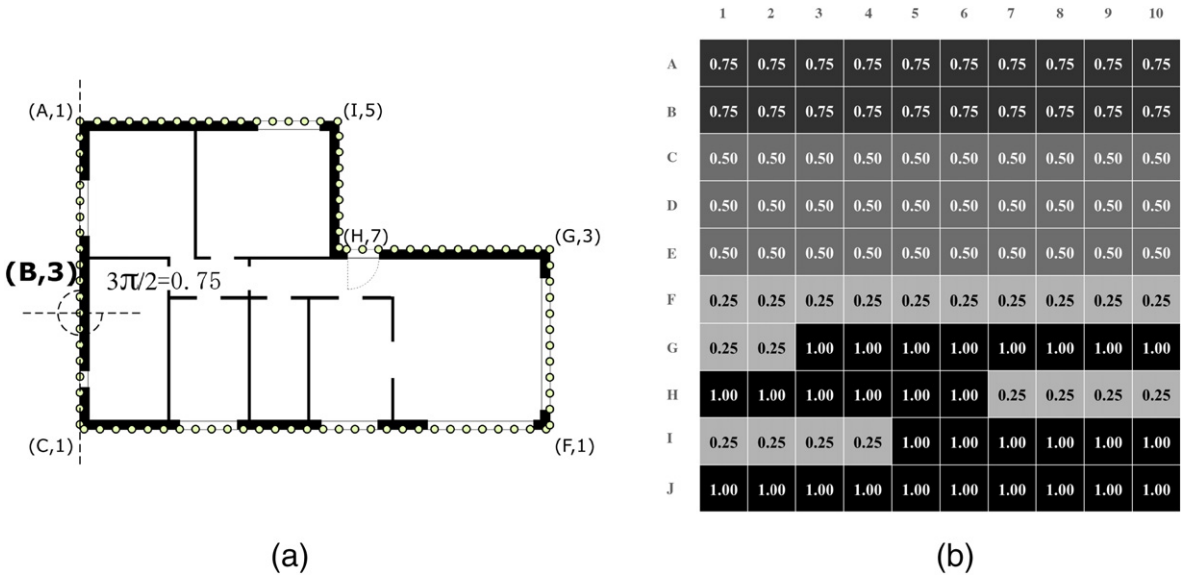
图2a展示了一个示例，其中转向函数的角度在点(B,3)处被测量，其值为$3\pi/2$，该描述子采用保持宽高比的变体。特征点从初始点(A,1)开始，该点是形状包围盒左上角最近的边界点，并以逆时针方向描绘形状轮廓。因此，各角度值在0至2π之间变化，随后经过归一化处理，使其取值范围为0至1。图2b则将转向函数（TF）描述子表示为一个梯度矩阵。由于平面布局呈正交结构，形状的边仅取四个可能的值：{$\pi/2$、π、$3\pi/2$、2π} ={0.25、0.50、0.75、1.00}。

2.1.3. Grid-Based (GB) descriptor

基于网格的描述子受Sajjanhar和Lu[19]的工作启发，其基本思想是将形状置于一个与平面图外墙体平行的正方形网格之下。对于网格中的每个单元格，其中心位置可能被形状区域占据（记为1），也可能未被占据（记为0）。该表示形式是一个二值向量，其长度等于网格单元格的总数，各元素值按照从左至右、从上至下的顺序读取。

图3a展示了一个以网格叠加的平面布局示例。在该示例中，点(B,8)的值为0，而点(F,9)的值为1，具体取决于平面布局中的区域是否位于该单元格的中心之下。

[1] The gradient matrix of the four representations is used only for visual comparison of different floor plans. The agglomerative hierarchical algorithm uses each data point as a 1-dimensional vector.



	1	2	3	4	5	6	7	8	9	10
A	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0.75
B	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0.75
C	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
D	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
E	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
F	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25
G	0.25	0.25	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H	1.00	1.00	1.00	1.00	1.00	1.00	0.25	0.25	0.25	0.25
I	0.25	0.25	0.25	0.25	1.00	1.00	1.00	1.00	1.00	1.00
J	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Fig. 2. Turning Function (TF) descriptor. (a) Example of the measuring angle in point (B,3) that has the value of 0.75, which corresponds to $3\pi/2$. The wall corners are marked with the matrix index to depict the counter-clockwise order of the feature points. (b) Vector in the form of a gradient matrix, where 0 is white and 1 is black, for angles ranging from 0 to 2π.

not. Fig. 3b represents the corresponding binary vector as a matrix. Each matrix entry has the corresponding value in the overlaid grid in the floor plan.

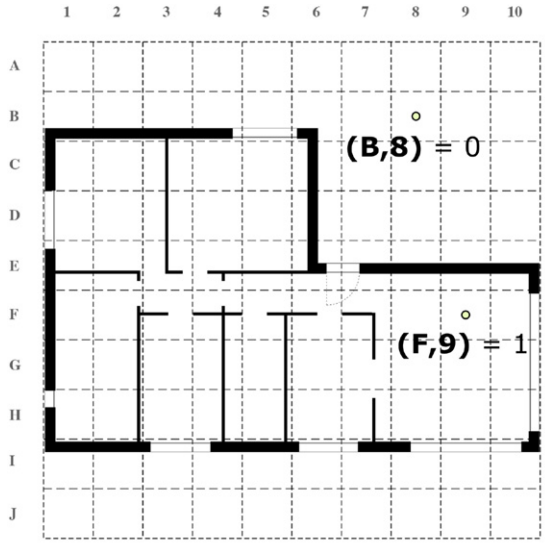
2.1.4. Tangent Distance (TD) descriptor

The Tangent Distance (TD) descriptor consists in determining the distance of a straight-line tangent to the shape contour to the bounding box centre. As floor plans are orthogonal shapes, ultimately the tangent line coincides with the exterior wall. The shape has its perimeter marked with points at regular length intervals starting on the nearest point on the shape perimeter to the top-left bounding rectangle. In every point, a straight line is drawn tangent to the shape and the distance is measured to the centre point. The vector has its values normalised—measured distance divided by the longest distance.

Fig. 4a depicts an example of the descriptor variant for preserved aspect ratio. The feature point (G,10) has a normalised distance value of 0.11 of its tangent to the centre. Fig. 4b illustrates the resulting vector in the form of a gradient matrix.

2.2. Clustering algorithm

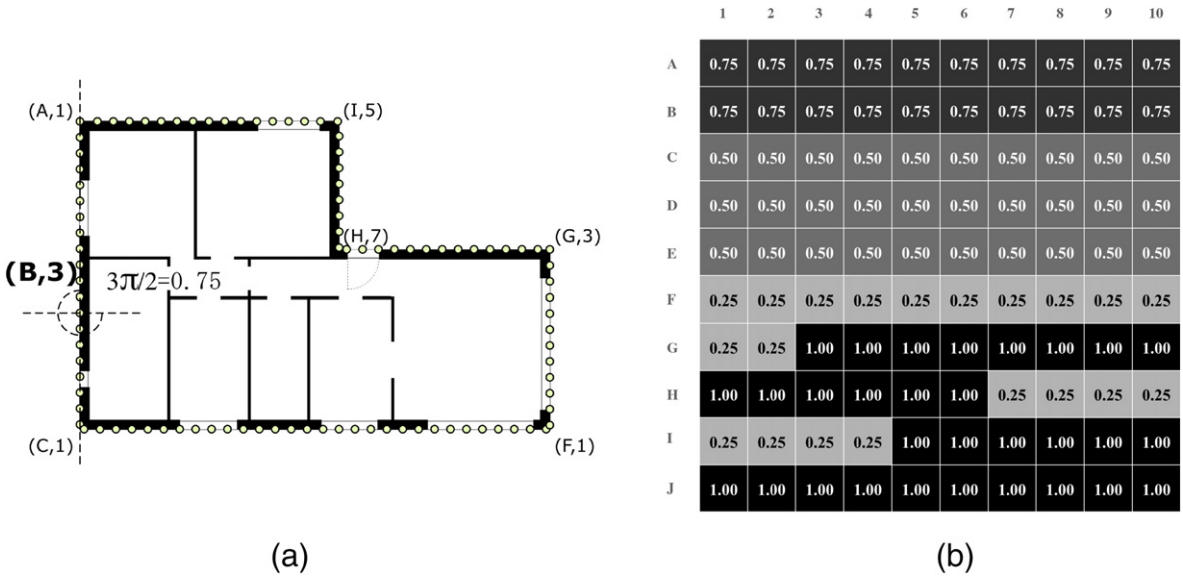
The dataset was clustered using an agglomerative hierarchical algorithm with Ward linkage [33] and the Euclidean distance as



	1	2	3	4	5	6	7	8	9	10
A	0	0	0	0	0	0	0	0	0	0
B	0	0	0	0	0	0	0	0	0	0
C	1	1	1	1	1	1	0	0	0	0
D	1	1	1	1	1	1	0	0	0	0
E	1	1	1	1	1	1	1	1	1	1
F	1	1	1	1	1	1	1	1	1	1
G	1	1	1	1	1	1	1	1	1	1
H	1	1	1	1	1	1	1	1	1	1
I	0	0	0	0	0	0	0	0	0	0
J	0	0	0	0	0	0	0	0	0	0

(b)

Fig. 3. Grid-Based (GB) descriptor. (a) Example of two point measurements. Point (B,8) is outside the floor plan area thus having the value of 0. Meanwhile, point (F,9) falls within the floor plan area and has a value of 1. (b) Vector in the form of a matrix (white is 0 and black is 1) depicting the corresponding cell value in the overlaid grid in the floor plan. Only the cell centre is used to measure the presence of the floor plan.



	1	2	3	4	5	6	7	8	9	10
A	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0.75
B	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0.75
C	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
D	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
E	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
F	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25
G	0.25	0.25	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H	1.00	1.00	1.00	1.00	1.00	1.00	0.25	0.25	0.25	0.25
I	0.25	0.25	0.25	0.25	1.00	1.00	1.00	1.00	1.00	1.00
J	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

图2. 转向函数（TF）描述符。（a）以点（B,3）为例，其测量角度值为0.75，对应于3π/2。墙体拐角处标注了矩阵索引，用于表示特征点的逆时针排列顺序。（b）以梯度矩阵形式呈现的向量，其中0为白色、1为黑色，对应的角度范围为0至2π。

否。图3b将相应的二值向量表示为一个矩阵，其中每个矩阵元素的值对应于平面图中叠加网格中的相应数值。

2.1.4. Tangent Distance (TD) descriptor

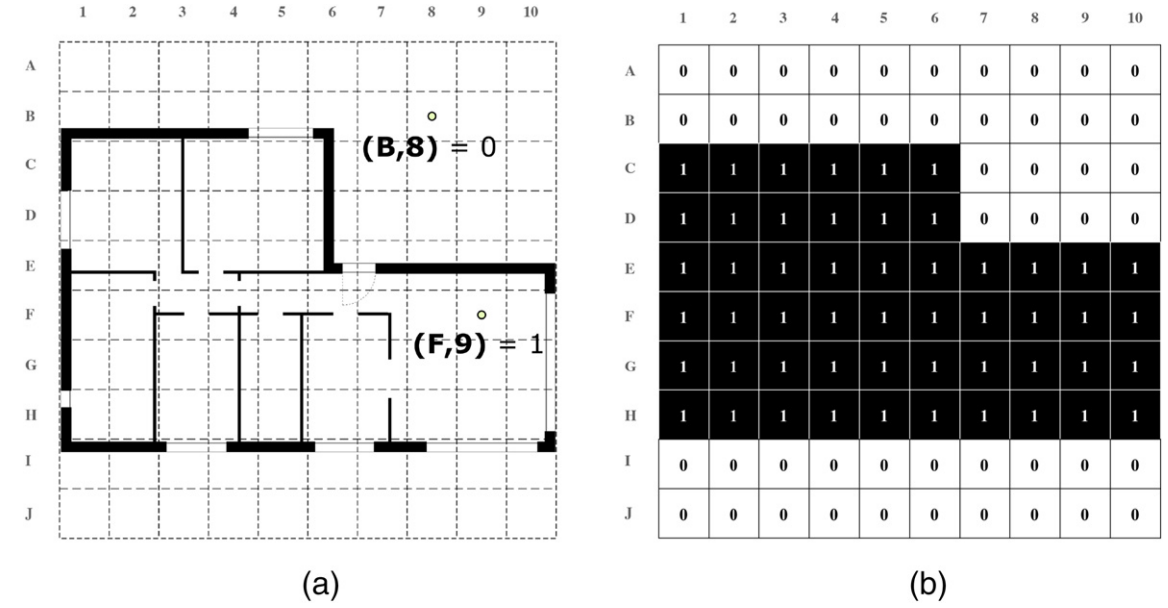
切线距离（TD）描述子通过计算与形状轮廓相切的直线到包围盒中心的距离来构建。由于平面图通常为正交形状，最终该切线与外墙重合。在形状的周界上，以从包围矩形左上角最近的点开始，按等间距标记若干点；在每个标记点处，均作一条与形状轮廓相切的直线。

形状及其到中心点的距离被测量。该向量的各分量经归一化处理——即测量得到的距离除以最大距离。

图4a展示了用于保持宽高比的描述符变体的一个示例。特征点(G,10)与其切线到中心的归一化距离值为0.11。图4b以梯度矩阵的形式展示了所得向量。

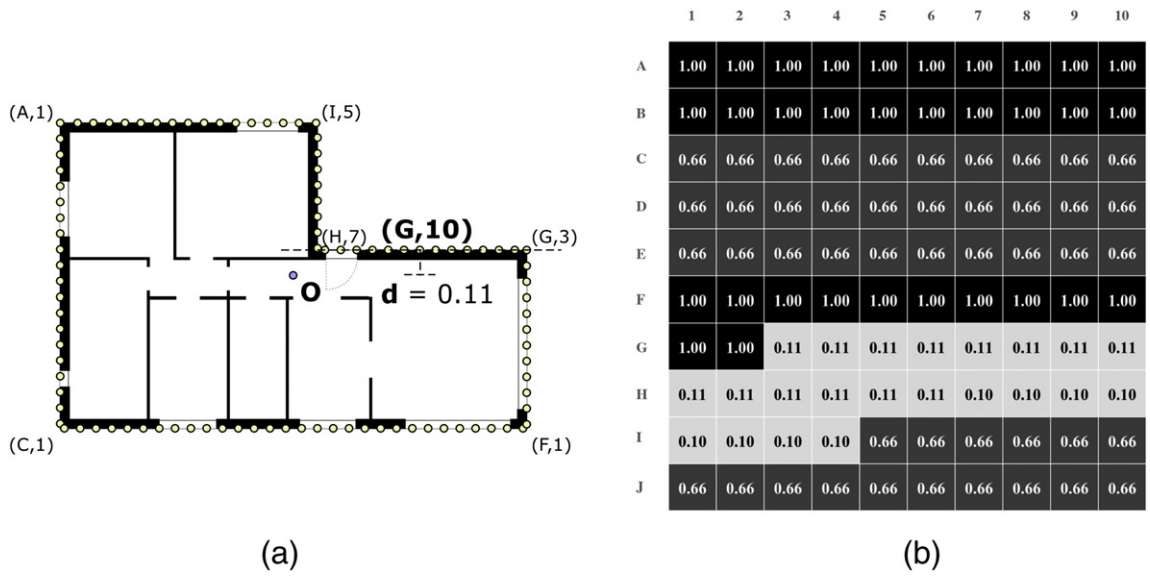
2.2. Clustering algorithm

该数据集采用基于Ward凝聚法的层次聚类算法进行聚类，距离度量采用欧氏距离。



	1	2	3	4	5	6	7	8	9	10
A	0	0	0	0	0	0	0	0	0	0
B	0	0	0	0	0	0	0	0	0	0
C	1	1	1	1	1	1	0	0	0	0
D	1	1	1	1	1	1	0	0	0	0
E	1	1	1	1	1	1	1	1	1	1
F	1	1	1	1	1	1	1	1	1	1
G	1	1	1	1	1	1	1	1	1	1
H	1	1	1	1	1	1	1	1	1	1
I	0	0	0	0	0	0	0	0	0	0
J	0	0	0	0	0	0	0	0	0	0

图3. 基于网格的描述符。（a）两个点测量示例。（B,8）点位于平面图区域之外，因此其值为0；而（F,9）点位于平面图区域内，其值为1。（b）以矩阵形式表示的向量（白色为0，黑色为1），刻画了叠加在平面图上的网格中各单元格的对应取值。仅采用单元格中心位置来判定平面图的存在与否。



	1	2	3	4	5	6	7	8	9	10
A	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
B	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
C	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.66
D	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.66
E	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.66
F	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
G	1.00	1.00	0.11	0.11	0.11	0.11	0.11	0.11	0.11	0.11
H	0.11	0.11	0.11	0.11	0.11	0.11	0.10	0.10	0.10	0.10
I	0.10	0.10	0.10	0.10	0.66	0.66	0.66	0.66	0.66	0.66
J	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.66

Fig. 4. Tangent Distance (TD) descriptor. (a) Example of the tangent distance for the feature point (G,10), which has the normalised distance value of 0.11. The horizontal walls corners are marked following the counter-clockwise order. (b) Vector in the form of a gradient matrix, where 0 is white and 1 is black.

the dissimilarity measure between different floor plan designs (feature vectors). Hierarchical clustering is based on the assumption that there is maximal quantifiable information when a set of elements is ungrouped, and that this information is captured by an objective function. In the case of agglomerative hierarchical clustering, the algorithm starts by considering as many clusters as the available data points and placing each data point in a cluster. It proceeds by merging two existing clusters that optimise an objective function. In this case the function is a variance criterion minimising the total within-cluster variance. At each step of the agglomerative process, the two clusters to be merged are dependent on the least increase in the total within-cluster variance. The process then proceeds iteratively until all clusters are grouped into a single global cluster.

Although the linkage criterion used in hierarchical clustering can be of different types, Ward's complete linkage aims to find compact clusters and was therefore preferred in this work. A similar linkage is the complete linkage clustering [41], where the distance between two different clusters is calculated by considering all pair-wise interactions between the elements in the two clusters. It then uses the distance of the pair of points that is farthest away from each other as the distance between the two clusters. It also aims to create compact clusters and to compute faster. For large populations it is an alternative to the Ward's criterion as it is faster. In this work, all results employed the Ward's criterion.

There are several measures available to determine the dissimilitude of two descriptor vectors [42]. In this work the dissimilitude between two feature vectors was calculated by the Euclidean distance for *N*-dimensions, with *N* being the length of the feature vector describing the floor plan design.

2.3. Synthetic dataset

The dataset of floor plan designs was created using a generative design algorithm, named the Evolutionary Program for the Space Allocation Problem (EPSAP) [34–36]. This algorithm combines an Evolution Strategy (ES) technique and a Stochastic Hill Climbing (SHC) method in a two-stage approach. The EPSAP is capable of generating multi-storey floor plans where parametric, non-rigid, and non-fixed vertical circulation elements evolve during the search process in interaction with the remaining spaces.

From a set of requirements defined by the user and given as input (see Subsection 3.1 for an example of the required input information), the generative design process initialises by creating, at the first ES generation, randomly distributed and dimensioned rectangles (each corresponding to a room) in the 2-dimensional plan—each storey has its own 2-dimensional plan. Each design solution is evaluated with a weighted sum of several objectives. These objectives are connectivity (interior doors), adjacency (proximity between rooms), room dimensions and area (according to minimum size of the smallest rectangle side and minimum floor area, respectively), compactness of the floor plan, room overflow in relation to a building boundary (when specified by the user), opening dimensions (to satisfy minimum width and window-to-floor ratio), and opening orientation (when specified by the user).

At every ES generation, the SHC method is called to randomly transform the different architectural elements in the floor plan (rooms, stairs, elevators, cluster of spaces, openings, walls, and the floor plans as a whole). The SHC method applies geometric actions such as translation, reflection, rotation, stretching, alignment of elements, permutation of element type, and changes to the element's orientation. The transformation action randomly selects the element, direction, and magnitude of change from the admissible geometric values. Then, the candidate solutions are evaluated. If the action produces an equal or better solution, the change is preserved, otherwise it is discarded. The SHC stage continues iteratively until reaching the SHC termination criterion—the difference between the moving average and the last iteration of the best individuals' average performance is greater than a defined threshold. Then, solutions having better performance than the average of the population are preserved for the next ES generation, while the remaining ones are discarded and substituted with new randomly generated ones, thus initiating a new ES cycle. When the ES termination criterion is reached, the algorithm stops and displays the results to the user.

As the EPSAP produces a large number of alternative floor plans, some kind of aggregation mechanism is required to help users compare and analyse the generated solutions. This is the motivation for the development of this study as described in Subsection 2.1.

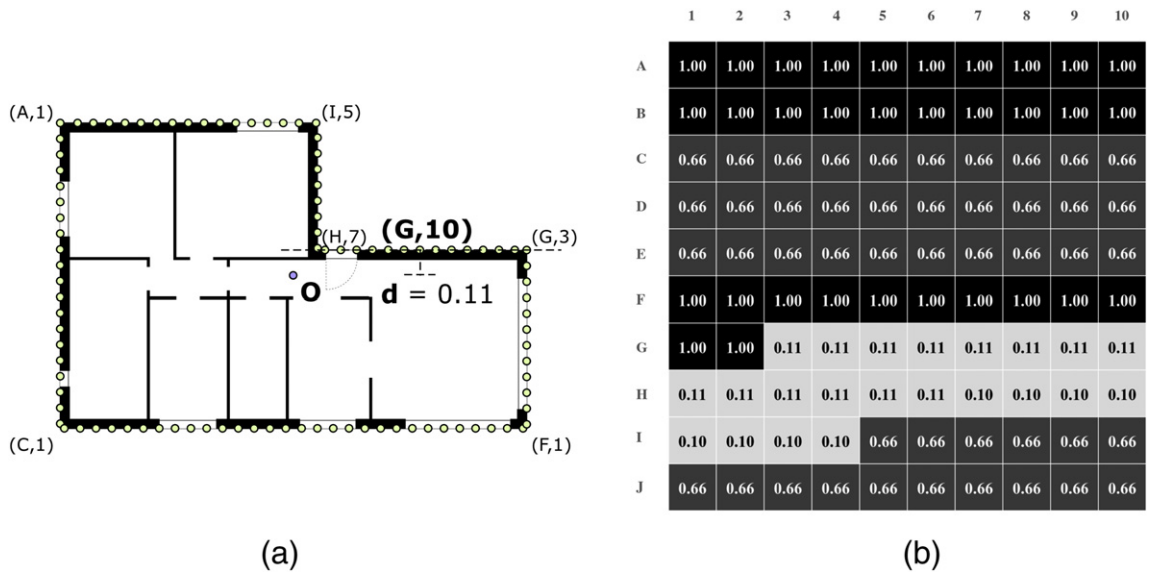


图4. 切线距离（TD）描述子。（a）特征点（G,10）的切线距离示例，其归一化距离值为0.11。水平墙面的角点按逆时针方向标记。（b）以梯度矩阵形式表示的向量，其中0表示白色，1表示黑色。

不同平面设计方案（特征向量）之间的相异性度量。层次聚类基于这样一个假设：当一组元素尚未分组时，其所包含的信息量达到最大，而这一信息可通过一个目标函数来刻画。在凝聚式层次聚类中，算法首先将每个数据点各自视为一个簇，随后通过合并两个能够使目标函数最优的现有簇来逐步进行聚类。在本例中，该目标函数为一种方差准则，旨在最小化簇内总方差。在凝聚过程的每一步，待合并的两个簇均由使簇内总方差增量最小的原则决定。这一过程 iteratively 持续进行，直至所有簇被归并为一个全局簇。

尽管层次聚类中使用的链接准则有多种类型，但Ward法的全链准则旨在寻找紧凑的簇，因此在本研究中被优先采用。与此类似的还有全链聚类[41]，该方法通过考虑两个簇中所有元素之间的两两交互来计算它们之间的距离，并以距离最远的一对点之间的距离作为这两个簇之间的距离。该方法同样旨在生成紧凑的簇，并且计算速度更快。对于大规模数据集，由于其计算效率更高，可作为Ward准则的替代方案。在本研究中，所有结果均采用了Ward准则。

目前已有多种方法可用于度量两个描述符向量之间的相异程度[42]。在本研究中，采用欧几里得距离来计算两个特征向量之间的相异度，其中N表示特征向量的维度数，N则为用于描述平面布局设计的特征向量的长度。

2.3. Synthetic dataset

平面布局设计数据集是通过一种名为“空间分配问题演化规划”（EPSAP）的生成式设计算法构建的[34–36]。该算法采用两阶段方法，将进化策略（ES）技术与随机爬山法（SHC）相结合。EPSAP能够生成包含参数化、非刚性构件的多层平面布局。

且在搜索过程中，非固定的垂直交通要素会与其余空间相互作用而不断演化。

给定由用户定义并作为输入的一组需求（有关所需输入信息的示例，请参见第3.1小节），生成式设计过程首先在第一代ES中，在二维平面内随机分布并确定尺寸的矩形（每个矩形对应一个房间）——每层楼均具有其独立的二维平面。每个设计方案均通过若干目标的加权求和进行评估。这些目标包括：连通性（室内门）、邻接关系（房间之间的接近程度）、房间尺寸与面积（分别依据最小矩形边长和最小 floor 面积要求）、平面紧凑性、房间超出建筑边界的情况（如用户有指定）、洞口尺寸（以满足最小宽度及窗地比要求），以及洞口朝向（如用户有指定）。

在每一代进化策略（ES）中，都会调用SHC方法，对平面布局中的不同建筑元素（房间、楼梯、电梯、空间群、洞口、墙体以及整个平面布局）进行随机变换。SHC方法施加的几何操作包括平移、反射、旋转、拉伸、元素对齐、元素类型的置换以及元素朝向的改变。变换操作会从允许的几何取值范围内随机选择要变换的元素、变换方向及变换幅度。随后，对候选解进行评估；若该操作产生的解优于或等于当前最优解，则保留该变更，否则予以舍弃。SHC阶段以迭代方式持续进行，直至满足SHC终止条件——即移动平均值与上一次迭代中最佳个体平均性能之间的差异超过预设阈值。此时，性能优于种群平均值的解将被保留用于下一代进化策略，其余解则被舍弃，并由新的随机生成解所替代，从而开启一个新的进化策略周期。当进化策略的终止条件被满足时，算法停止运行，并将结果呈现给用户。

由于EPSAP能够生成大量备选的平面布局方案，因此需要某种聚合机制来帮助用户对所生成的方案进行比较与分析。这正是本研究在第2.1节中所述开发工作的主要动因。

3. Results

3.1. Showcase specifications

A single-family three-bedroom house was used as an illustrative example. In addition to the three bedrooms (R_{6-8}), a hall (R_1), a kitchen (R_2), a living room (R_3), a corridor (R_5), and two bathrooms (R_4 and R_9) were specified. Topologically, all spaces have connection to the hall or the corridor. The kitchen also has an interior door connecting to the living room. One of the bathrooms serves the public area of the house and the other is connected to the corridor of the private part of the house, which is connected to all bedrooms. The interior connectivity (M_{con}) is defined in Matrix (1), where 1 represents an interior door connecting two rooms and 0 indicates the absence of doors connecting them.

$$M_{con} = \begin{array}{c} \\ R_1 \\ R_2 \\ R_3 \\ R_4 \\ R_5 \\ R_6 \\ R_7 \\ R_8 \\ R_9 \end{array} \begin{array}{c} \begin{array}{ccccccccc} R_1 & R_2 & R_3 & R_4 & R_5 & R_6 & R_7 & R_8 & R_9 \end{array} \\ \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{pmatrix} \end{array} \quad (1)$$

All interior doors must have 0.90 m wide except the living room doors, which are 1.40 m. With the exception of the horizontal circulation spaces and one of the bathrooms, all remaining spaces have at least one window (the living room has two). The hall has one exterior door facing north (orientation up). No other topological requirement was added, such as opening orientation or space location on the floor plan.

The detailed showcase requirements are presented in Table 1, where the information relating to each room is listed. These include space name (M_{sn}), space function type (M_{st}, where 0 represents circulation spaces, 1 rooms, and 2 kitchens and bathrooms), minimum floor side dimension (M_{fd}), minimum floor area (M_{fa}), exterior opening width (M_{eow}) and height (M_{eoh}), space window-to-floor ratio (M_{wfr}), clear area in the outside of opening (M_{eoa}), exterior opening orientation (M_{eoo}), and interior doors minimum width (M_{idw}). The thicknesses of walls are 0.32 m for the exterior wall (t_{ew}) and 0.11 m for the interior wall (t_{iw}). The floor plan design (FPD) must have a construction area inferior to 200 m^2 (a_c).

Using these requirements as input, the EPSAP algorithm ran a single time to generate 72 alternative floor plans from a population of 576 individuals (each individual is a candidate solution). The generative design process took 136 s in a 2.8 GHz Quad-core computer with 8 GB of RAM. Multi-threading was used. The floor plans improved over a total of 1790 iterations by minimising penalties for not satisfying the user specifications. The best individual had a fitness of 98,265.1 in the first iteration and 2.2 in the last iteration, which resulted from not attaining the aimed floor plan area.

3.2. Clustering results

As the purpose of this work was to provide the EPSAP algorithm with clustering capabilities to help the user deal with a large number of generated solutions, and because the type of shapes and resulting numbers are not known a priori, an unsupervised clustering approach was used. That is, the number of clusters does not depend on the real number of different shapes in the generated set but on the number of alternative solutions that the user wants or might analyse. As the complexity of the floor plans increases, the number of alternative shapes also grows, easily reaching numbers that become intractable for the decision-maker. The clustering mechanism is independent from the number of clusters and the number of floor plan designs, thus may be scaled up or down only affecting computation time. As the vector in every clustering process had the same length (100 values), the type of shape representation did not affect the performance of the algorithm. However, the results had significant differences depending on the shape descriptor.

During the preparatory work, a survey was conducted to determine which clustering features human experts use to group floor plans [31,32]. The survey analysis determined the main features, such as shape and indoor room arrangement. However, human experts are generally inconsistent during the clustering process—for instance, the same individual may sometimes gather floor plans by shape and in other times by indoor space arrangement. This resulted in having groups where a floor plan A has similar shape as a floor plan B and the latter has the same internal arrangement as a floor plan C. However, C has no similarity whatsoever with A, despite the three being in the same cluster. Therefore, the results of the survey were not used as a ground truth due to this changing behaviour. As an alternative, a reference clustering was determined by typifying shapes from designs found in the dataset. Fig. 5 depicts such partition (labelled from **A'** to **I'**) with the typified shape on the left of each group letter.

There is the O-shape, four rotated L-shapes, two rotated T-shapes, and two reflected Z-shapes. Group **A'** (O-shape) has 7 designs; **B'**

Table 1
Case study specifications for spaces and openings.

Storey	Space					Ext. opening					Int. door
		M_{sn}	M_{st}	M_{fd}	M_{fa}	M_{eow}	M_{eoh}	M_{wfr}	M_{eoa}	M_{eoo}	M_{idw}
L_1	R_1	Hall	0	1.40 m	5.0 m^2	1.20 m	2.00 m		{1.80 m, 3.00 m}	North	0.90 m
	R_2	Kitchen	2	2.60 m	15.0 m^2		1.00 m	0.1	{3.00 m, 3.00 m}		0.90 m
	R_3	Living room	1	4.00 m	20.0 m^2	{5.00 m, 4.00 m}	{2.40 m, 2.40 m}		{3.00 m, 3.00 m}		1.40 m
	R_4	Bathroom	2	1.80 m	3.0 m^2						0.90 m
	R_5	Corridor	0	1.40 m	3.0 m^2						0.90 m
	R_6	Bedroom	1	3.50 m	18.0 m^2		1.00 m	0.1	{3.00 m, 3.00 m}		0.90 m
	R_7	Bedroom	1	3.00 m	15.0 m^2		1.00 m	0.1	{3.00 m, 3.00 m}		0.90 m
	R_8	Bedroom	1	2.70 m	12.0 m^2		1.00 m	0.1	{3.00 m, 3.00 m}		0.90 m
	R_9	Priv. bathroom	2	1.80 m	3.0 m^2	0.60 m	0.60 m		{3.00 m, 3.00 m}		0.90 m

$t_{ew} = 0.32\text{m}, t_{iw} = 0.11\text{m}, \text{and} a_c \leq 200\text{m}^2$.

3. 结果

3.1. Showcase specifications

以一栋单户三居室住宅作为示例。除三个卧室（R_{6-8}）外，还设有一间门厅（R_1）、一间厨房（R_2）、一间起居室（R_3）、一条走廊（R_5）以及两间卫生间（R_4和R_9）。在拓扑结构上，所有空间均与门厅或走廊相连；厨房还设有一扇通向起居室的内门。其中一间卫生间服务于住宅的公共区域，另一间则与住宅私密部分的走廊相连，而该走廊又与所有卧室相通。内部连通性（M_{con}）由矩阵（1）定义，其中“1”表示连接两间的内门存在，“0”则表示两者之间无门相连。

$$M_{con} = \begin{array}{c} \\ R_1 \\ R_2 \\ R_3 \\ R_4 \\ R_5 \\ R_6 \\ R_7 \\ R_8 \\ R_9 \end{array} \begin{array}{c} \begin{array}{ccccccccc} R_1 & R_2 & R_3 & R_4 & R_5 & R_6 & R_7 & R_8 & R_9 \end{array} \\ \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{pmatrix} \end{array} \quad (1)$$

所有室内门的宽度均为0.90米，唯独客厅门为1.40米。除水平交通空间及其中一间卫生间外，其余所有房间均至少设有一扇窗户（客厅设有两扇）。走廊设有一扇朝北的外门（朝向为上）。此外未附加其他拓扑约束条件，如开启方向或房间在平面图中的位置。

详细的展示要求如表1所示，其中列出了各房间的相关信息，包括空间名称（M_{sn}）、空间功能类型（M_{st}，其中0表示交通空间，1表示居室，2表示厨房和卫生间）、地面最小边长（M_{fd}）、地面最小面积（M_{fa}）、外窗洞口宽度（M_{eow}）和高度（M_{eoh}）、窗地比（M_{wfr}）、洞口外侧净面积（M_{eoa}）、外窗朝向（M_{eoo}）以及室内门最小宽度（M_{idw}）。外墙厚度为0.32 m（t_{ew}），并且内墙的厚度为0.11 m（t_{iw}）。平面布局设计（FPD）的建筑面积必须小于200 m^2（a_c）。

以这些需求作为输入，EPSAP算法运行一次，从包含576个个体的种群中生成了72种备选平面布局方案（每个个体即一个候选解）。在一台配备8 GB内存、主频为2.8 GHz的四核计算机上，该生成式设计过程耗时136秒，并采用了多线程技术。在总共1790次迭代过程中，通过最小化未满足用户规范的惩罚值，各平面布局方案逐步得到优化。在首次迭代中，最优个体的适应度评分为98,265.1；而在最后一次迭代中，其适应度降至2.2，这是由于未能达到目标平面布局面积所致。

3.2. Clustering results

鉴于本研究旨在为EPSAP算法赋予聚类能力，以帮助用户处理大量生成的方案，并且由于形状类型及相应数量在事前未知，因此采用了无监督聚类方法。也就是说，聚类数目并不取决于所生成方案集中实际存在的不同形状数量，而取决于用户希望或可能分析的备选方案数量。随着平面布局复杂度的增加，备选形状的数量也随之增长，极易达到决策者难以处理的程度。该聚类机制与聚类数目及平面设计方案数量无关，因而可通过调整规模来适应不同的需求，仅影响计算时间。由于每次聚类过程中的向量长度均相同（100个数值），形状表示方式本身并不影响算法的性能。然而，基于不同的形状描述子，所得结果存在显著差异。

在前期准备工作中，开展了一项调查，以确定人类专家在对平面图进行聚类时所依据的特征[31,32]。调查分析明确了主要特征，如形状和室内房间布局。然而，在聚类过程中，人类专家往往表现出不一致性——例如，同一位专家有时会按形状对平面图进行归类，而另一些时候则按室内空间布局进行归类。这导致出现这样的情况：某一平面图A的形状与另一平面图B相似，而后者又与某一平面图C的内部布局相同，但尽管三者处于同一簇中，C却与A毫无相似之处。因此，鉴于这种行为的多变性，调查结果并未被用作真值。作为替代方案，通过归纳数据集中设计的形状类型，确定了一个参考聚类。图5展示了这一划分（标注为A’至I’），其中每组字母左侧标示了相应的典型形状。

其中包括一个O形、四个旋转的L形、两个旋转的T形以及两个反射的Z形。A’组（O形）包含7种设计；B’

表1 空间与开口的案例研究规格。

Storey	Space					Ext. opening					Int. door
		M_{sn}	M_{st}	M_{fd}	M_{fa}	M_{eow}	M_{eoh}	M_{wfr}	M_{eoa}	M_{eoo}	M_{idw}
L_1	R_1	Hall	0	1.40 m	5.0 m^2	1.20 m	2.00 m		{1.80 m, 3.00 m}	North	0.90 m
	R_2	Kitchen	2	2.60 m	15.0 m^2		1.00 m	0.1	{3.00 m, 3.00 m}		0.90 m
	R_3	Living room	1	4.00 m	20.0 m^2	{5.00 m, 4.00 m}	{2.40 m, 2.40 m}		{3.00 m, 3.00 m}		1.40 m
	R_4	Bathroom	2	1.80 m	3.0 m^2						0.90 m
	R_5	Corridor	0	1.40 m	3.0 m^2						0.90 m
	R_6	Bedroom	1	3.50 m	18.0 m^2		1.00 m	0.1	{3.00 m, 3.00 m}		0.90 m
	R_7	Bedroom	1	3.00 m	15.0 m^2		1.00 m	0.1	{3.00 m, 3.00 m}		0.90 m
	R_8	Bedroom	1	2.70 m	12.0 m^2		1.00 m	0.1	{3.00 m, 3.00 m}		0.90 m
	R_9	Priv. bathroom	2	1.80 m	3.0 m^2	0.60 m	0.60 m		{3.00 m, 3.00 m}		0.90 m

$t_{ew} = 0.32$米，$t_{iw} = 0.11$米，以及$a_c \leq 200$米^2。

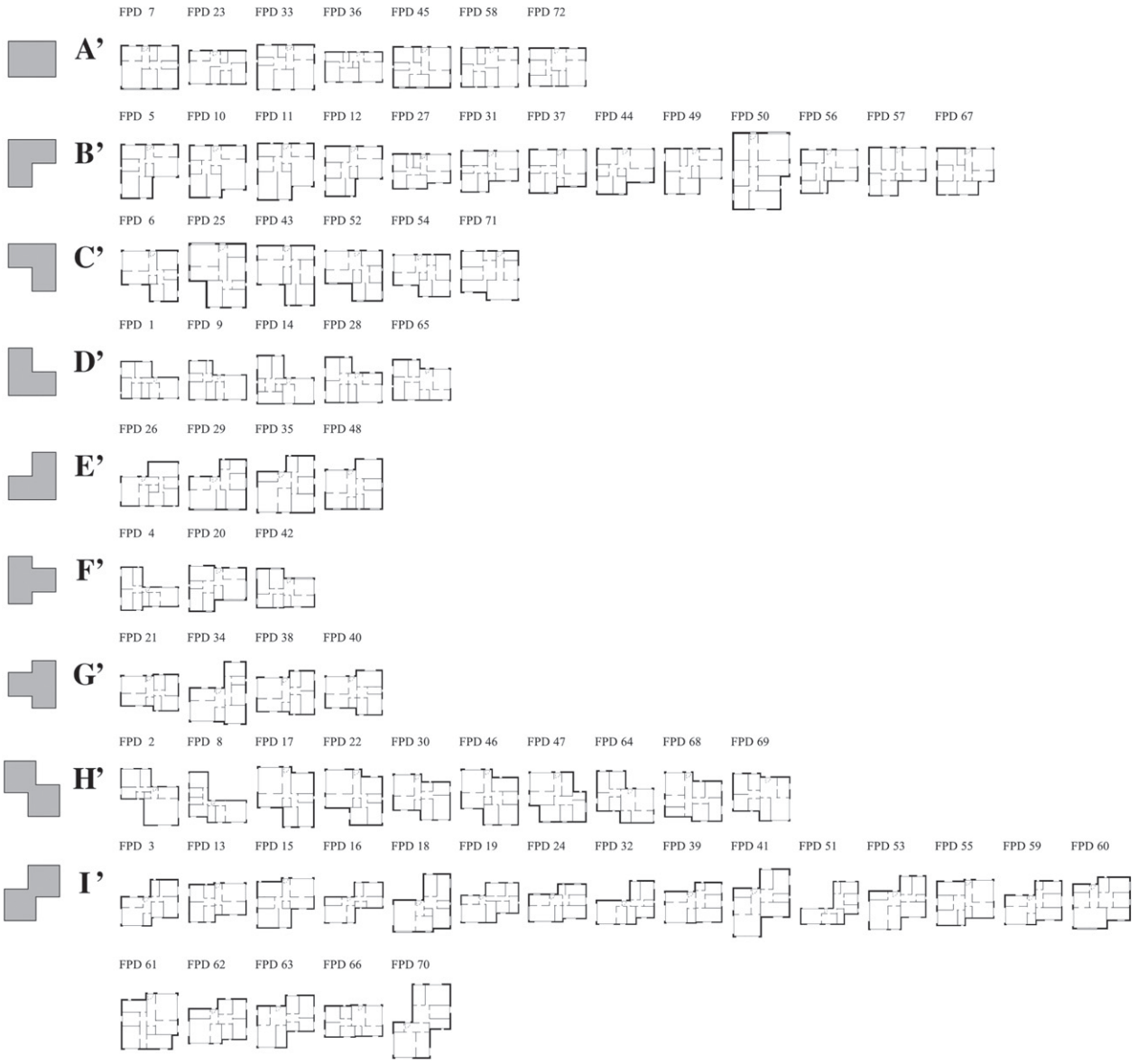


Fig. 5. Reference clustering and shape type by group.

(top-left L-shape) has 13; **C'** (top-right L-shape) has 6; **D'** (L-shape) has 5; **E'** (reflected L-shape) has 4; **F'** (rotated left T-shape) has 3; **G'** (rotated right T-shape) has 4; **H'** (Z-shape) has 10; and, finally, **I'** (reflected Z-shape) has 20 designs.

Several measures have been proposed to determine the quality of the resulting groups and comparing those clusters with a reference group of the data. The measures of comparison have to be able to handle minor data perturbations as well as missing data, but remain sensitive enough when two clustering methods produce different results from the same data [43]. In Rand [43] an index is proposed that is based on a measure of similarity between two different clusterings of the same dataset and considers how each pair of data points is assigned in each clustering. If the pair of points i,j is placed together—assigned to the same cluster—in both clusterings, or if they are placed in different clusters in both clusterings, this is considered a similarity trait between the two clusterings. The dissimilarity is observed when the pair of points is placed together in one clustering and separated in the other [43]. Therefore, for any two clusterings Y, Y' of N points $X_1, X_2, \ldots, X_N$, the similarity between them is calculated by Eq. (2), where $\gamma_{ij} = 1$ if the pair of points i,j appears in both clusterings in the same relation and $\gamma_{ij} = 0$ if the pair of points does not have the same kind of relations in both clusterings.

$$c(Y, Y') = \sum_{i<j}^{N} \frac{\gamma_{ij}}{\binom{N}{2}} \tag{2}$$

Additionally, each descriptor (and its alternative variant of non-fixed aspect ratio) was evaluated according to the perceptual coherence of each group and between groups. A group is considered coherent if it presents a dominant shape (the shape that appears the highest number of times in a group) with a lower number of outlier designs. Confusion matrices are used to compare descriptor variants. These are presented in a table format where two clusterings from the same dataset can be compared by showing the number of elements that belong to the clusters of both clusterings, in each table

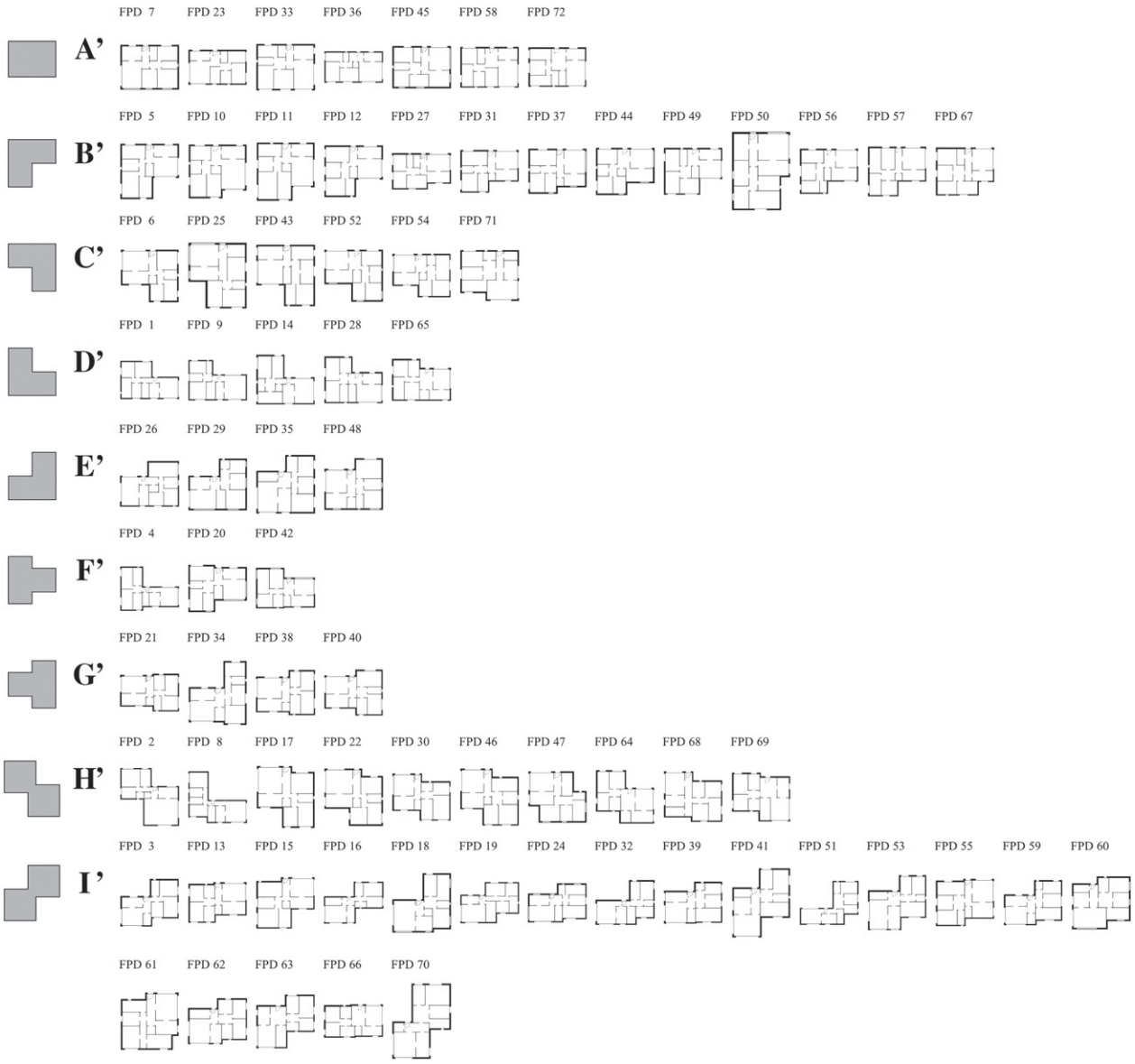


图5. 按组别进行的参考聚类与形状类型。

（左上角的L形）包含13种；C′（右上角的L形）包含6种；D′（L形）包含5种；E′（镜像L形）包含4种；F′（左旋T形）包含3种；G′（右旋T形）包含4种；H′（Z形）包含10种；最后，I′（镜像Z形）包含20种设计。

已提出多种指标用于评估所得聚类的质量，并将这些聚类与数据的参考聚类进行比较。这些比较指标既要能够应对数据的轻微扰动以及缺失数据，又要在两种聚类方法基于同一数据产生不同结果时保持足够的敏感性[43]。Rand[43]提出了一种基于同一数据集上两种不同聚类之间相似性度量的指数，该指数考察每对数据点在两种聚类中的分配情况。若某对数据点i、j在两种聚类中均被分配至同一簇，或均被分配至不同簇，则将其视为两种聚类间的一种相似特征；而当这对数据点在一种聚类中被分配至同一簇，但在另一种聚类中被分配至不同簇时，则被视为一种相异特征[43]。因此，对于任意两种聚类 Y，Y' 个 N 点中的 X_1、X_2、…和 X_N，它们之间的相似性通过式（2）计算得到，其中，若点对 i、j 在两种聚类中以相同的关系出现，则 γ_{ij} = 取值为 1；若该点对在两种聚类中不具有相同类型的关系，则 γ_{ij} = 取值为 0。

$$c(Y,Y') = \sum_{i<j}^{N} \frac{\gamma_{ij}}{\binom{N}{2}} \tag{2}$$

此外，每个描述子（及其非固定宽高比的替代变体）均根据各组内及组间的感知一致性进行了评估。当一个组中存在一种占主导地位的形状（即在该组中出现次数最多的形状），且异常设计的数量较少时，该组被视为具有一致性。混淆矩阵被用于比较不同描述子变体；这些矩阵以表格形式呈现，通过展示同时属于两个聚类划分中相应簇的元素数量，从而对来自同一数据集的两种聚类结果进行对比。

entry. These are usually used to compare a clustering predicted by a machine learning algorithm and a clustering that is a reference clustering. The columns and rows represent each group for the two descriptors.

3.2.1. Point Distance (PD) descriptor results

For PD descriptor, Fig. 6 depicts the clustering results (for fixed aspect ratio) and the group's dominant shape at left of the group letter. The group outliers were placed at the end of each group row for readability.

This descriptor presents six unique dominant shapes from a total of nine possible ones, none of the groups was free from outliers, clustering accuracy of 70.83%, and Rand index of 0.861. The number of designs per group varies between 4 and 14. The group with the highest number of dominant shape designs (N_d) was group **D** with 9 and the groups with the lowest number of outliers were **D**, **G**, **H**, and **I** with one. Outliers exist in all groups.

From a perceptual analysis, when compared to the reference clustering partition, the PD descriptor is unable to have a fully coherent group. For instance, group **A** has the L-shape as the dominant shape the type and FPD 4, 8, 42, and 64 as outliers. Group **B** follows the Z-shape type and has as outliers FPD 6, 25, 43, 52, 54, and 71, which would fit better in the top-right L-shape (dominant shape absent from this partition). Group **C** only has 2 outliers (FPD 26 and 38) and has a reflected Z-shape. The top-left L-shape group **D** has only 1 outlier (FPD 20). Group **E** aggregates the O-shape type and have 2 outliers (FPD 27 and 37) that would fit in group **D**. Groups **F** and **H** have the same reflected Z-shape type as group **C** and only have one incorrectly assigned design (FPD 50 and 34, respectively). Finally, the last group **I** has a reflected L-shape with one outlier (FPD 61).

Table 2a presents the confusion matrix of this fixed aspect ratio descriptor variant against the reference clustering partition. It is noticeable that designs in partitions **B'** and **I'** are dispersed over four or more groups of the descriptor results, thus showing the difficulty of the PD descriptor in correctly determining the top-left L-shape and the reflected Z-shape types. It is also observable that the top-right L-shape (partition **C'**), rotated left T-shape (**F'**), and rotated right T-shape (**G'**) are outliers in several descriptor groups (**B**; **A** and **D**; and **C**, **F**, and **H**, respectively).

Comparing the fixed aspect ratio variant of this descriptor with the non-fixed one (see Fig. A.10 in Appendix A), the performance decreases with an clustering accuracy (Ac) to 66.67% and Rand index (R_i) to 0.852. Despite having one group with no outlier (group **C**) and finding the same number of unique shape groups (see Table 2b), the descriptor with this feature loses accuracy in groups **B**, **E**, **G**, **H**, and **I**; however, it improves in groups **C** and **D** (see Table 2c).



Fig. 6. Clustering results using Point Distance (PD) descriptor.

条目。这些指标通常用于比较由机器学习算法预测得到的聚类与作为参考的聚类。表中的列和行分别对应两种描述符的各个簇。

3.2.1. Point Distance (PD) descriptor results

对于PD描述符，图6展示了聚类结果（在固定长宽比条件下），并在各组字母的左侧标注了该组的主导形状。为便于阅读，各组中的离群点被放置在该组行的末尾。

该描述符从总共九种可能的形状中识别出六种独特的主导形状，所有聚类均存在异常点；聚类准确率为70.83%，Rand指数为0.861。各组的设计数量介于4至14之间。拥有最多主导形状设计的组（N_d）为D组，包含9个设计；异常点数量最少的组为D、G、H和I，每组仅含1个异常点。所有组均存在异常点。

从感知分析的角度来看，与参考聚类划分相比，PD描述子无法形成一个完全一致的簇。例如，簇A以L形为主要形状，而FPD 4、8、42和64则被视为离群点；簇B以Z形为主要形状，其离群点为FPD 6、25、43、52、54和71，这些样本更适于归入右上角的L形簇（该簇缺乏主导形状）。

在该分区中，C组仅包含2个异常点（FPD 26和38），且呈现镜像Z形分布。位于左上角的L形分布组D仅包含1个异常点（FPD 20）。E组归为O形类型，并包含2个异常点（FPD 27和37），这两个点本应归属于D组。F组和H组与C组具有相同的镜像Z形分布特征，各自仅存在1个被错误归类的设计方案（分别为FPD 50和34）。最后，I组呈现镜像L形分布，且包含1个异常点（FPD 61）。

表2a展示了该固定宽高比描述子变体与参考聚类划分之间的混淆矩阵。值得注意的是，B’和I’两类中的设计模式分散于描述子结果的四个或更多类别中，这表明PD描述子在正确识别左上角L形以及反射Z形类型方面存在困难。此外，还可以观察到，右上角L形（C’）、左旋T形（F’）以及右旋T形（G’）分别在多个描述子类别中表现为离群点（分别为B、A与D，以及C、F与H）。

将该描述子的固定宽高比变体与非固定宽高比变体进行比较（参见附录A中的图A.10），聚类准确率（*Ac*）下降至66.67%，Rand指数（R_i）降至0.852。尽管其中一个簇（簇C）不含异常点，且识别出的唯一形状簇数量相同（参见表2b），但带有此特征的描述子在簇B、E、G、H和I上的精度有所降低；然而，在簇C和D上其精度有所提升（参见表2c）。

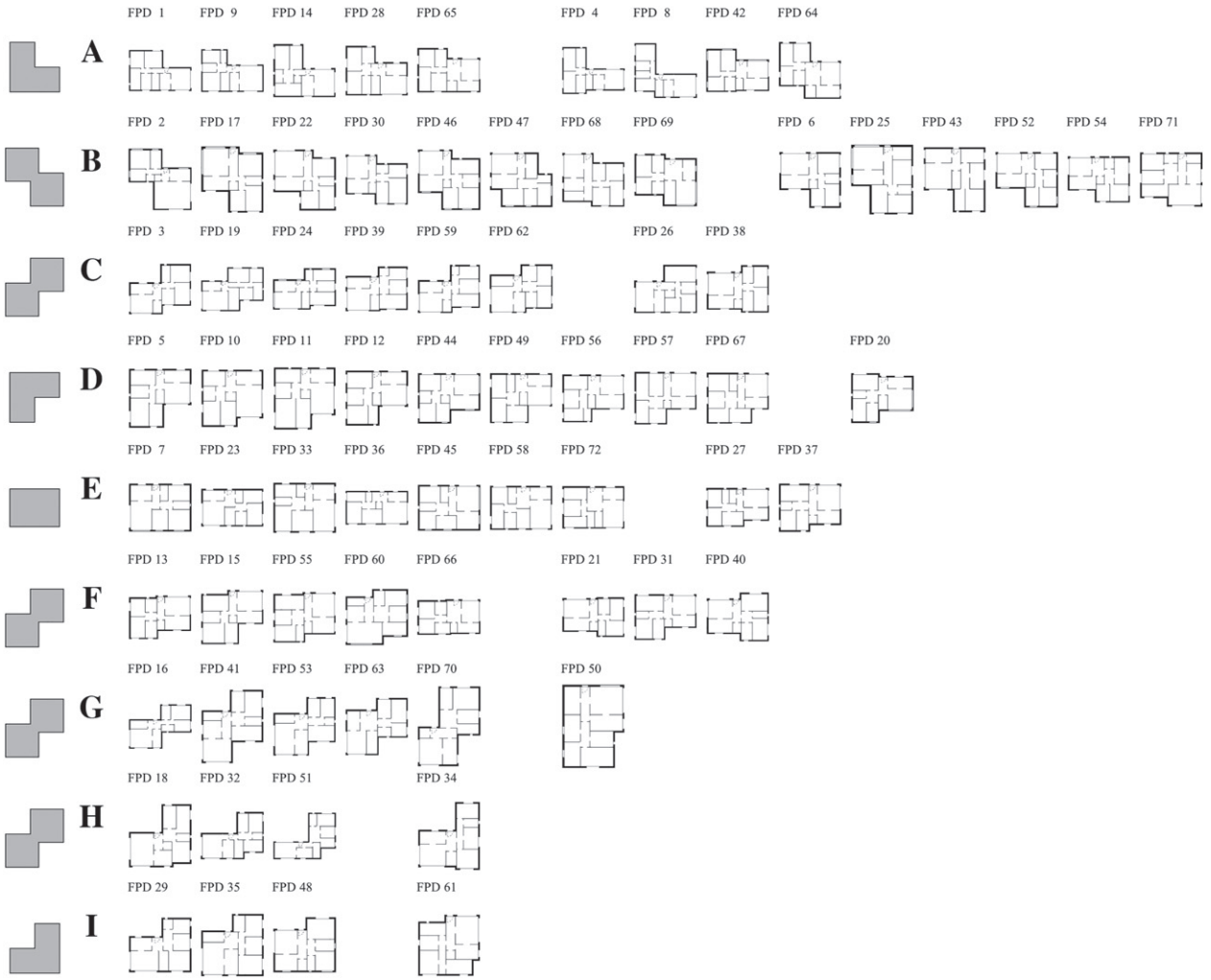


图6. 使用点距离（PD）描述子的聚类结果。

Table 2
Point Distance (PD) confusion matrices.

(a)

Reference clustering \ Fixed aspect ratio	A	B	C	D	E	F	G	H	I
A'					7				
B'				9	2	1	1		
C'		6							
D'	5								
E'			1						3
F'	2			1					
G'			1			2		1	
H'	2	8							
I'			6			5	5	3	1

(b)

Reference clustering \ Non-fixed aspect ratio	A	B	C	D	E	F	G	H	I
A'					7				
B'				10	1	1			1
C'		4			2				
D'	5								
E'							1	3	
F'	2			1					
G'						1	2	1	
H'	2	5			3				
I'			7			4	5	2	2

(c)

Fixed aspect ratio \ Non-fixed aspect ratio	A	B	C	D	E	F	G	H	I
A	9								
B		9			5				
C			2			2	4		
D				10					
E				1	8				
F						4	4		
G			3						3
H			2					2	
I								4	

3.2.2. Turning Function (TF) descriptor results

Fig. 7 presents the results for the TF descriptor and the dominant shape in each group. The TF descriptor has 6 unique shape groups (N_u), 2 groups without any outlier (N_o), clustering accuracy of 66.67%, and Rand index of 0.842 (R_i). The number of designs per group varies between 4 and 15. The groups with the highest number of dominant shape designs (N_d) were **C** and **D** with 8. The groups with no outliers were **D** and **H** (N_e).

The perceptual analysis of the group coherence shows that group **A** has two outliers (FPD 4 and 8) and the dominant shape type is the L-shape. Group **B** follows the Z-shape and has FPD 28, 42, and 65 incorrectly assigned. **C** has a reflected Z-shape type and the largest number of outliers (FPD 21, 26, 29, 35, 38, 40, and 48) that mix reflected L-shape and rotated right T-shape types. Group **D** has no outliers and its shape type is the top-left L-shape. Group **E** dominant shape is the top-right L-shape with 4 outliers (FPD 17, 22, 46, and 69)



Fig. 7. Clustering results using Turning Function (TF) descriptor.

表2 点间距（PD）混淆矩阵。

Reference clustering	Fixed aspect ratio								
	A	B	C	D	E	F	G	H	I
A'					7				
B'				9	2	1	1		
C'		6							
D'	5								
E'			1						3
F'	2			1					
G'			1			2		1	
H'	2	8							
I'			6			5	5	3	1

(a)

Reference clustering	Non-fixed aspect ratio								
	A	B	C	D	E	F	G	H	I
A'					7				
B'				10	1	1			1
C'		4			2				
D'	5								
E'							1	3	
F'	2			1					
G'						1	2	1	
H'	2	5			3				
I'			7			4	5	2	2

(b)

Fixed aspect ratio	Non-fixed aspect ratio								
	A	B	C	D	E	F	G	H	I
A	9								
B		9			5				
C			2			2	4		
D				10					
E				1	8				
F						4	4		
G			3						3
H			2					2	
I								4	

(c)

3.2.2. Turning Function (TF) descriptor results

图7展示了TF描述符及各组中占主导地位的形状特征结果。TF描述符共识别出6个具有唯一形状特征的组（N_u），其中2个组未包含任何异常值（N_o），聚类准确率为66.67%，Rand指数为0.842（R_i）。各组的设计数量在4至15之间不等。主导形状设计数量最多的组为C组和D组，均为8个（N_d）；未出现异常值的组为D组和H组（N_e）。

对群体一致性的感知分析表明，A组存在两个离群点（FPD 4和8），其主导形状类型为L形。B组的主导形状类型为Z形，并且FPD 28、42和65被错误地归类。C组的形状类型为反向Z形，且离群点数量最多（FPD 21、26、29、35、38、40和48），这些离群点同时包含反向L形和右旋T形两种类型。D组无离群点，其形状类型为左上L形。E组的主导形状类型为右上L形，存在4个离群点（FPD 17、22、46和69）。



图7. 基于转向函数（TF）描述符的聚类结果。

whose shape fits in group **B** with Z-shape type. The O-shape group is **F** and has 6 outliers (FPD 20, 27, 37, 47, 50, and 56). Groups **G**, **H**, and **I** have the same dominant shape as **C** (reflected Z-shape). **G** only has 1 outlier (FPD 31, a top-left L-shape) and **I** has 2 outliers (FPD 51 and 34).

Table 3a compares the fixed aspect ratio descriptor variant with the reference clustering partition. The designs in partitions **B'**, **F'**, **H'**, and **I'** are spread over three or more groups, thus indicating the TF descriptor's difficulty in correctly capturing the shape top-left L-shape, rotated left T-shape, Z-shape, and reflect Z-shape types, respectively. One may also note that shapes from partitions **E'**, **F'**, and **G'** were unable to dominate any group.

When considering the non-fixed aspect ratio descriptor variant (results are depicted in Fig. A.13 in Appendix A), the performance of *Ac* increases to 69.44% and the R_i to 0.858. One of the two groups that had no outliers is also lost. Table 3b shows the increase of clustering accuracy for shapes in partitions **B'**, **D'**, and **F'** and decreases in **C'** and **E'**. When comparing both descriptor variants in Table 3c, group **I** has the largest shift of designs, capturing 8 that were previously in group **C**. The groups that acquire designs from other groups are **A**, **C**, **D**, **F**, and **H**.

3.2.3. Grid-Based (GB) descriptor results

Fig. 8 illustrates the GB descriptor clustering. GB only identifies 5 unique shape groups (N_u) and one group was free from outliers (N_o). The clustering accuracy and Rand index were the lowest of all descriptors with only 55.56% (*Ac*) and 0.824 (R_i), respectively. The number of designs per group varies between 4 and 12. The groups with the highest number of dominant shape designs (N_d) were **C** and **G** with 8. Group **F** had no outliers (N_e). Group **I** has two dominant shapes.

GB descriptor has the lowest group coherence of all the descriptors' results. For example, groups **A** and **I** have more outliers than dominant shapes—**A** (O-shape type) has FPD 1, 9, 21, 24, 27, 42, and 66 as outliers, and **B** has FPD 38, 40, and 48, and one of the two sets FPD 52, 54, and 71 (top-right L-shape) or FPD 30, 47, 69 (Z-shape). The Z-shape groups **B** and **E** have 4 (FPD 4, 14, 28, and 65) and 2 outliers (FPD 6 and 43). Groups **C**, **D**, and **H** have as dominant shape the reflected Z-shape type and has dissimilar designs FPD 26 and 29, FPD 5, 10, 11, 34, and 35, and FPD 25 and 50, respectively. Group **G**, with top-left L-shape type, has FPD 13, 15, 20, and 55 presents differing designs.

The confusion matrix, depicted in Table 4a for fixed aspect ratio, shows designs dispersed over all groups, forming heterogeneous partitions. For instance, reference clustering partitions **B'** and **I'** have designs distributed over four or more descriptor groups—**A**, **D**, **G**, and **H**, and **A**, **C**, **D**, **G**, and **H**, respectively. Therefore, the fixed aspect ratio variant of this descriptor cannot accurately capture the differences between all shapes.

However, if allowed to change the design aspect ratio, the GB descriptor significantly improves its accuracy, reaching 75.00% for *Ac* (the highest of all descriptors) and 0.874 for R_i. The group designs are depicted in Fig. A.12 in Appendix A. It also achieves 7 unique shape groups (N_u) and two groups without any outlier (N_o). Table 4b shows the performance improvement in all groups as dominant shape designs increase in all partitions. The comparison of the two descriptor variants in Table 4c illustrates how designs that initially were in group **A** are now assigned to groups **A** to **F**. Other examples are the new groups **B**, **C**, **D**, and **E**, which capture designs that were assigned to several groups.

3.2.4. Tangent Distance (TD) descriptor results

The results from the TD descriptor are displayed in Fig. 9. Out of all the descriptors and variants in this study, the TD descriptor presents the best results. It was able to determine 6 unique shape groups (N_u; similar to PD and TF descriptors) and only 1 group had no outliers. The clustering accuracy and Rand index were the highest of the fixed aspect ratios descriptors variant with 73.61% and 0.873 (R_i), respectively. The number of designs per cluster varies between 5 and 14. The group with the highest number of dominant shapes was **D** with 10 and the lowest number of outliers was group **C** with none.

This descriptor has the highest group coherence of all. However, there are still outliers. For instance, group **A** has the L-shape as the dominant shape type but also captures 4 outliers (FPD 4, 8, 42, and 64), three of those due to small recesses in the bottom wall. It is observable that FPD 64 clearly belongs to the Z-shape type group. Group **B** has 6 outliers (FPD 6, 25, 43, 52, 54, and 71)—all fitting the top-right L-shape instead of the dominant Z-shape type. Top-left L-shape in group **D** has a single outlier (FPD 20), which fits the rotated left T-shape due to a small recess in the top wall. For similar reasons, group **E** with O-shape type has FPD 27 (top-left L-shape) as an outlier. Groups **F** and **G** have the same reflected Z-shape type. The outliers of these groups are FPD 21 and 31 and outlier FPD 50, respectively. Despite having the same shape type, TD descriptor partitioned designs into two groups because the concave turns in the walls have different size segments. Group **H** has 2 outliers (FPD 18 and 34) in the dominant shape type reflected L-shape. Once again, the descriptor did not consider these designs with a different shape despite the small recess in the bottom wall. Finally, the last group **I**,

Table 3
Turning Function (TF) confusion matrices.

(a)

Reference clustering \ Fixed aspect ratio	A	B	C	D	E	F	G	H	I
A'						7			
B'				8		4	1		
C'					6				
D'	3	2							
E'			4						
F'	1	1				1			
G'			3						1
H'	1	4			4	1			
I'			8				4	5	3

(b)

Reference clustering \ Non-fixed aspect ratio	A	B	C	D	E	F	G	H	I
A'						7			
B'				9		2	1	1	
C'					5	1			
D'	5								
E'			3						1
F'	2			1					
G'			1						3
H'	1	4			4	1			
I'			5				3	8	4

(c)

Fixed aspect ratio \ Non-fixed aspect ratio	A	B	C	D	E	F	G	H	I
A	5								
B	3	4							
C			5					2	8
D				8					
E					9	1			
F				2		10		1	
G							4	1	
H								5	
I			4						

其形状归入B组，属于Z形类型。O组为F组，包含6个异常值（FPD 20、27、37、47、50和56）。G、H和I组的主导形状与C组相同（均为反向Z形）。G组仅含1个异常值（FPD 31，左上角L形），而I组则有2个异常值（FPD 51和34）。

表3a将固定长宽比描述子变体与参考聚类划分进行了比较。分区B’、F’、H’和I’中的设计分散于三个或更多簇中，这表明该TF描述子在分别准确刻画左上L形、左旋T形、Z形以及镜像Z形等形状类型方面存在困难。此外，还可注意到，来自分区E’、F’和G’的形状均未能主导任何一个簇。

当考虑非固定宽高比描述子变体时（结果见附录A中的图A.13），Ac的性能提升至69.44%，而R_i则提升至0.858。同时，原本无异常值的两个聚类组之一也消失了。表3b显示，B’、D’和F’分区中形状的聚类准确率有所提高，而在C’和E’分区中则有所下降。在表3c中比较两种描述子变体时，I组的设计转移最为显著，新增了8个此前位于C组的设计。从其他组获取设计的组别包括A、C、D、F和H。

3.2.3. Grid-Based (GB) descriptor results

图8展示了GB描述符的聚类结果。GB仅识别出5个独特的形状类别（N_u），且其中一个类别未包含异常值（N_o）。其聚类准确率和Rand指数均为所有描述符中最低，分别为55.56%（Ac）和0.824（R_i）。各类别所含设计数量介于4至12之间。具有最多主导形状设计的类别（N_d）为C和G，各有8个；类别F未见异常值（N_e）；类别I则包含两种主导形状。

GB描述符在所有描述符的结果中具有最低的组内一致性。例如，A组和I组中的异常值数量多于主导形状——A组（O形）的FPD 1、9、21、24、27、42和66为异常值，B组的FPD 38、40和48为异常值，而在FPD 52、54和71（右上L形）与FPD 30、47、69（Z形）这两组中也各有一组存在异常值。Z形的B组和E组分别有4个（FPD 4、14、28和65）和2个异常值（FPD 6和43）。C、D和H组以反向Z形为主要形状，但各自包含设计迥异的FPD：分别为FPD 26和29、FPD 5、10、11、34和35，以及FPD 25和50。G组以左上L形为主，其FPD 13、15、20和55呈现出不同的设计特征。

如表4a所示，针对固定宽高比的混淆矩阵显示，设计方案分散于所有类别中，形成了异质性划分。例如，参考聚类划分B’和I’具有设计分布于四个或更多的描述符组——A、D、G 和 H，以及 A、C、D、G 和 H，分别。因此，该描述符的固定宽高比变体无法准确捕捉所有形状之间的差异。

然而，若允许调整设计的宽高比，GB描述符的准确率显著提升，在所有描述符中达到最高水平，为75.00%；其AUC值亦增至0.874。各分组的设计如附录A图A.12所示。此外，该描述符还能够识别出7个独特的形状分组，并且其中两个分组中不存在任何异常样本。表4b显示，随着各分区中占主导地位的形状设计数量增加，所有分组的性能均有所改善。表4c中两种描述符变体的对比结果表明，原本归于A类的设计现已被分配至A至F共六个分组；此外，B、C、D和E等新分组则涵盖了此前被分配至多个分组的设计。

3.2.4. Tangent Distance (TD) descriptor results

TD描述符的分析结果如图9所示。在本研究涉及的所有描述符及其变体中，TD描述符的表现最佳。它能够识别出6个独特的形状类别（N_u；与PD和TF描述符类似），且仅有一个类别不含异常值。在固定宽高比的描述符变体中，其聚类准确率和Rand指数分别达到最高值，分别为73.61%和0.873（R_i）。各簇中的设计数量介于5至14之间。其中，具有最多主导形状的类别为D，包含10个设计；而异常值最少的类别为C，未发现任何异常值。

该描述符的组内一致性最高。然而，仍存在若干离群点。例如，A组以L形为主要形状类型，但也包含了4个离群点（FPD 4、8、42和64），其中3个是由于底壁上的小型凹槽所致。可以观察到，FPD 64显然属于Z形类型的组别。B组包含6个离群点（FPD 6、25、43、52、54和71），这些设计均符合右上L形，而非该组的主要Z形类型。D组中的左上L形仅有一个离群点（FPD 20），其因顶壁上的小型凹槽而呈现旋转后的左T形。出于类似原因，E组以O形为主，但FPD 27（左上L形）被视为离群点。F组和G组具有相同的反向Z形类型，其离群点分别为FPD 21、31以及FPD 50。尽管形状类型相同，TD描述符仍将这些设计划分为两个组别，这是因为各壁面的凹转处具有不同尺寸的分段。H组在其主要形状类型——反向L形中包含2个离群点（FPD 18和34）。同样，该描述符未将这些设计归为同一类别，尽管其底壁存在小型凹槽。最后，I组，

表3 转向功能（TF）混淆矩阵。

(a)

Reference clustering \ Fixed aspect ratio	A	B	C	D	E	F	G	H	I
A’						7			
B’				8		4	1		
C’					6				
D’	3	2							
E’			4						
F’	1	1				1			
G’			3						1
H’	1	4			4	1			
I’			8				4	5	3

(b)

Reference clustering \ Non-fixed aspect ratio	A	B	C	D	E	F	G	H	I
A’						7			
B’				9		2	1	1	
C’					5	1			
D’	5								
E’			3						1
F’	2			1					
G’			1						3
H’	1	4			4	1			
I’			5				3	8	4

(c)

Fixed aspect ratio \ Non-fixed aspect ratio	A	B	C	D	E	F	G	H	I
A	5								
B	3	4							
C			5					2	8
D				8					
E					9	1			
F				2		10		1	
G							4	1	
H								5	
I			4						

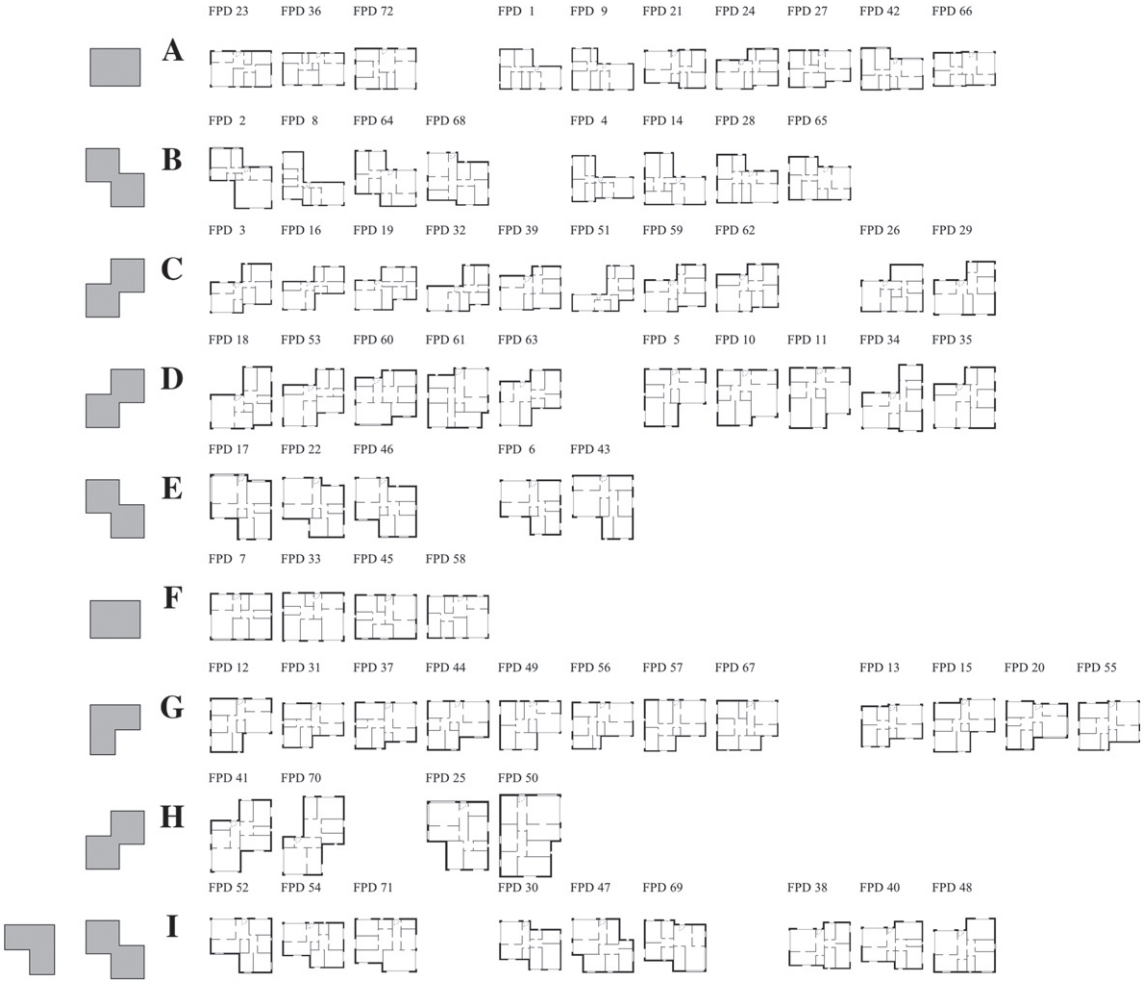


Fig. 8. Clustering results using Grid-Based (GB) descriptor.

with reflected Z-shape, has 2 outliers (FPD 38 and 40 with rotated right T-shape type).

Table 5a presents the confusion matrix for this descriptor against the reference clustering. Partition **A'** designs are fully included in group **E**. However, partition **B'** has three of its designs spread over three groups **E** to **G**, but the remaining 10 designs are assigned to group **D**. Partitions **C'**, **D'**, and **E'** are also assigned to a corresponding group—**B**, **A**, and **H**, respectively. Designs in partitions **G'** and **H'** are distributed over three (**F**, **H**, and **I**) and two groups (**A** and **B**). Finally, the largest reference clustering partition **I'** had its designs assigned to five groups (**C**, and **F** to **I**).

When considering the non-fixed aspect ratio descriptor variant (Fig. A.13 in Appendix A), the descriptor underperforms slightly in the clustering accuracy, which decreases to 72.22%, but improves in the Rand index to 0.876. Reference clustering partitions **B'** and **G'** are better partitioned in this descriptor variant, but accuracy is lost for

Table 4
Grid-Based (GB) confusion matrices.

Reference clustering	Fixed aspect ratio A	B	C	D	E	F	G	H	I
A'	3					4			
B'	1			3			8	1	
C'					2			1	3
D'	2	3							
E'			2	1					1
F'	1	1					1		
G'	1			1					2
H'		4			3				3
I'	2		8	5			3	2	

(a)

Reference clustering	Non-fixed aspect ratio A	B	C	D	E	F	G	H	I
A'						7			
B'				8			5		
C'					6				
D'	4	1							
E'									4
F'	1	1		1					
G'			1		3				
H'	1	5			1			3	
I'			12	6		1			1

(b)

Fixed aspect ratio	Non-fixed aspect ratio A	B	C	D	E	F	G	H	I
A	2	1	1	1	1	4			
B	4	4							
C			6	1					3
D			4	5					1
E					2			3	
F						4			
G				7			5		
H			2	1	1				
I		2			6				1

(c)

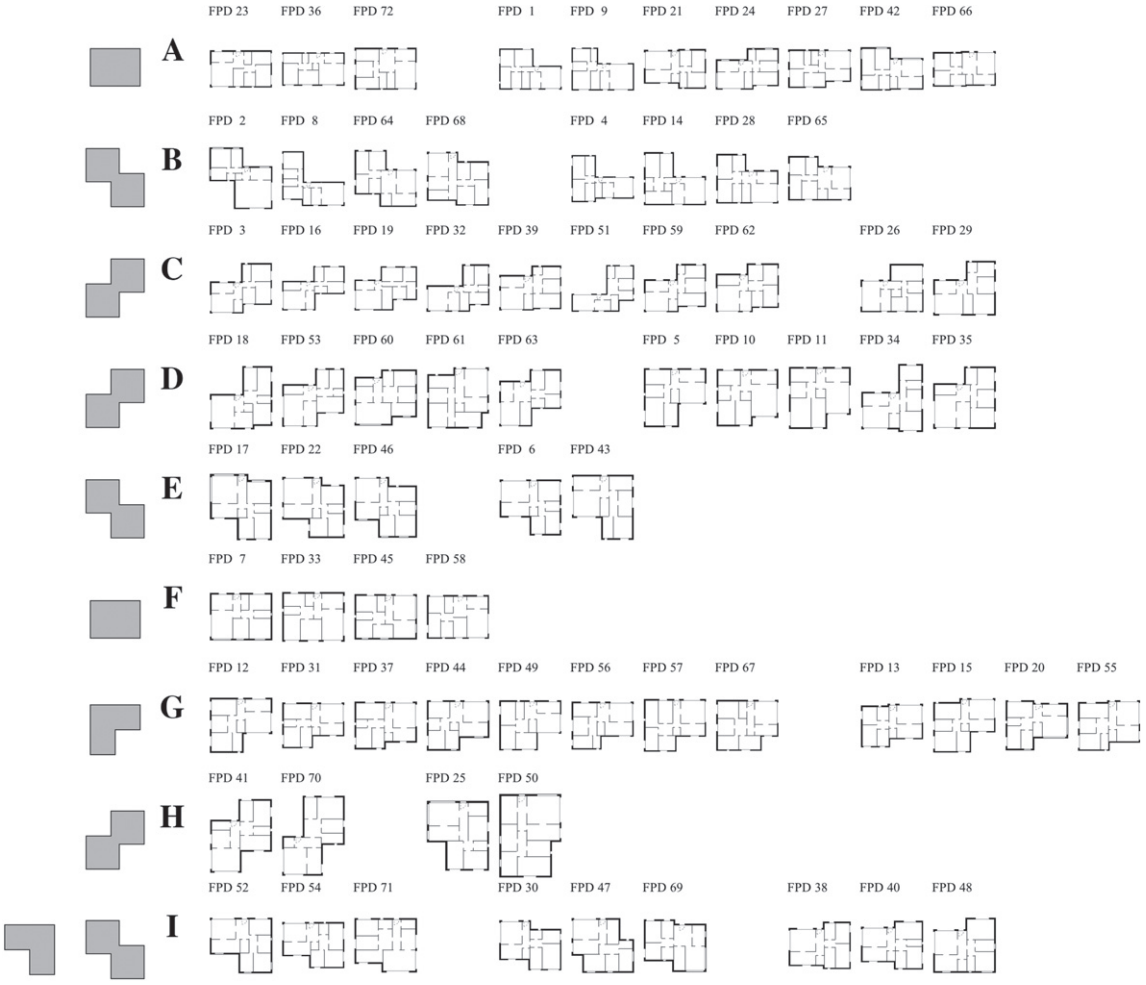


图8. 基于网格（GB）描述符的聚类结果。

呈反向Z形，其中存在2个异常值（FPD 38和40，为右旋T形）。

表5a展示了该描述符与参考聚类之间的混淆矩阵。A’分区的设计完全归属于E组。然而，B’分区中有3个设计分散于E至G三个组，而其余10个设计则被分配至D组。C’、D’和E’分区也分别被分配至相应的B、A和H组。G’和H’分区中的设计则分布于三个组别（F、H 和 I）以及两个组别（A 和 B）。最后，规模最大的参考聚类划分 I’ 将其设计分配至五个组别（C 以及 F 至 I）。

在考虑非固定宽高比描述子变体时（见附录A中的图A.13），该描述子在聚类准确率方面表现略逊，降至72.22%，但在Rand指数方面有所提升，达到0.876。在此描述子变体中，参考聚类划分B’和G’的划分效果更好，但准确性有所损失。

表4 基于网格（GB）的混淆矩阵。

Reference clustering	Fixed aspect ratio A	B	C	D	E	F	G	H	I
A’	3					4			
B’	1			3			8	1	
C’					2			1	3
D’	2	3							
E’			2	1					1
F’	1	1					1		
G’	1			1					2
H’		4			3				3
I’	2		8	5			3	2	

(a)

Reference clustering	Non-fixed aspect ratio A	B	C	D	E	F	G	H	I
A’						7			
B’				8			5		
C’					6				
D’	4	1							
E’									4
F’	1	1		1					
G’			1		3				
H’	1	5			1			3	
I’			12	6		1			1

(b)

Fixed aspect ratio	Non-fixed aspect ratio A	B	C	D	E	F	G	H	I
A	2	1	1	1	1	4			
B	4	4							
C			6	1					3
D			4	5					1
E					2			3	
F						4			
G				7			5		
H			2	1	1				
I		2			6				1

(c)

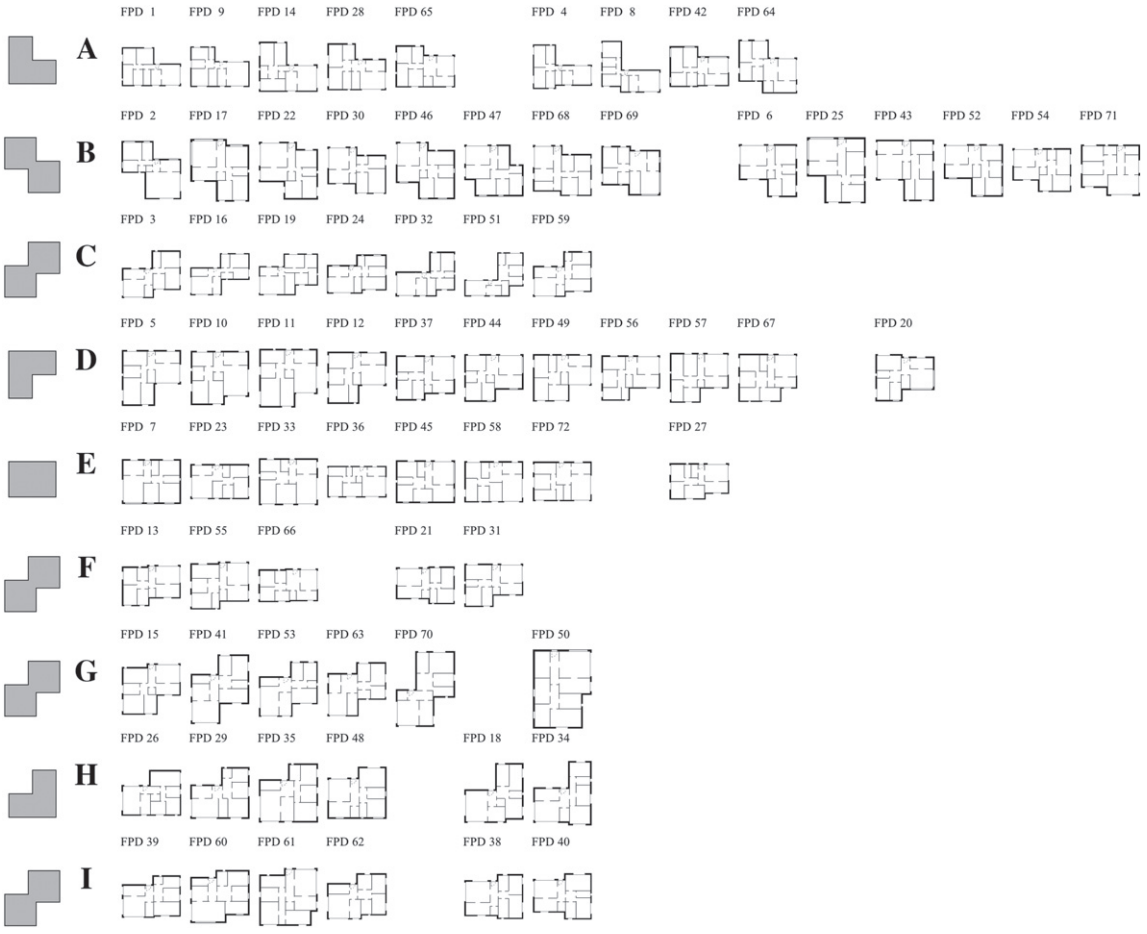


Fig. 9. Clustering results using Tangent Distance (TD) descriptor.

partitions **C'**, **E'**, **G'**, **H'**, and **I'** (Table 5b). Comparing both descriptor variants (Table 5c) groups **B**, **C**, and **E** to **I** have a few designs that have been shifted to other groups.

4. Discussion

Table 6 summarises per descriptor the number of unique shapes (N_u; number of groups with unique shape type), number of groups without outliers (N_o), the percentage of clustering accuracy (*Ac*; number of dominant shape designs per total of floor plan designs), and Rand index (R_i). It also lists the number of dominant shapes (N_d) and the number of outliers (N_e) per group. The descriptor with better R_i is Tangent Distance (TD) with 0.873 and 0.876 for fixed and non-fixed aspect ratio variants, respectively. However, Grid-Based (GB) presents the highest number of unique shape groups (N_u) and the highest *Ac* of 75% for the non-fixed aspect ratio descriptor variant.

The presence of outliers (N_e) in the Point Distance (PD) descriptor may indicate why some groups have designs dispersed by other

Table 5
Tangent Distance (TD) confusion matrices.

Reference clustering \ Fixed aspect ratio	A	B	C	D	E	F	G	H	I
A'					7				
B'				10	1	1	1		
C'		6							
D'	5								
E'								4	
F'	2			1					
G'						1		1	2
H'	2	8							
I'			7			3	5	1	4

(a)

Reference clustering \ Non-fixed aspect ratio	A	B	C	D	E	F	G	H	I
A'					7				
B'				12		1			
C'		5						1	
D'	5								
E'							2		2
F'	2			1					
G'							1	3	
H'	3	7							
I'			7		1	4	3	1	4

(b)

Fixed aspect ratio \ Non-fixed aspect ratio	A	B	C	D	E	F	G	H	I
A	9								
B	1	12						1	
C			3				2		2
D				11					
E				1	7				
F						1	3		1
G			4	1		1			
H								4		2
I							1		3	2

(c)



图9. 使用切线距离（TD）描述子的聚类结果。

分区C’、E’、G’、H’和I’（表5b）。比较两种描述符变体（表5c）后发现，B、C以及E至I组中存在若干设计方案已被调整至其他组别。

4. 讨论

表6按各描述符汇总了唯一形状的数量（N_u；具有唯一形状类型的簇数）、无异常值的簇数（N_o）以及聚类准确率的百分比（Ac）；每个平面布局设计总数中占主导地位的形状设计数量）以及兰德指数（R_i）。此外，还列出了各组中占主导地位的形状数量（N_d）和异常值数量（N_e）。在描述符性能方面，切线距离（TD）表现更优，其固定宽高比变体和非固定宽高比变体的得分分别为0.873和0.876。然而，基于网格的描述符（GB）在非固定宽高比变体中展现出最多的独特形状类别数量（N_u），且其得分最高，达到75%（Ac）。

点间距离（PD）描述符中存在异常值（N_e），这可能解释了为何某些组的设计与其他组相比呈现分散分布。

表5 切线距离（TD）混淆矩阵。

Reference clustering	Fixed aspect ratio								
	A	B	C	D	E	F	G	H	I
A’					7				
B’				10	1	1	1		
C’		6							
D’	5								
E’								4	
F’	2			1					
G’						1		1	2
H’	2	8							
I’			7			3	5	1	4

(a)

Reference clustering	Non-fixed aspect ratio								
	A	B	C	D	E	F	G	H	I
A’					7				
B’				12		1			
C’		5						1	
D’	5								
E’							2		2
F’	2			1					
G’							1	3	
H’	3	7							
I’			7		1	4	3	1	4

(b)

Fixed aspect ratio	Non-fixed aspect ratio								
	A	B	C	D	E	F	G	H	I
A	9								
B	1	12						1	
C			3				2		2
D				11					
E				1	7				
F					1	3		1	
G			4	1		1			
H							4		2
I						1		3	2

(c)

Table 6
Descriptors performance.

			Group																			
			A		**B**		**C**		**D**		**E**		**F**		**G**		**H**		**I**			
Descriptor	N_u	N_o	N_d	N_e	N_d	N_e	N_d	N_e	N_d	N_e	N_d	N_e	N_d	N_e	N_d	N_e	N_d	N_e	N_d	N_e	Ac	R_i
(a) Fixed aspect ratio																						
Point Distance (PD)	**6**	0	5	4	8	6	6	2	9	1	7	2	5	3	5	1	3	1	3	1	70.83%	0.861
Turning Function (TF)	**6**	**2**	3	2	4	3	8	7	8	0	6	4	7	6	4	1	5	0	3	1	66.67%	0.842
Grid-Based (GB)	5	1	3	7	4	4	8	2	5	5	3	2	4	0	8	4	2	2	3	6	55.56%	0.824
Tangent Distance (TD)	**6**	1	5	4	8	6	7	0	10	1	7	1	3	2	5	1	4	2	4	2	**73.61**%	**0.873**
(b) Non-fixed aspect ratio																						
Point Distance (PD)	6	1	5	4	5	4	7	0	10	1	7	6	4	2	5	3	3	3	2	1	66.67%	0.852
Turning Function (TF)	6	1	5	3	4	0	5	4	9	1	5	4	7	4	3	1	8	1	4	4	69.44%	0.858
Grid-Based (GB)	**7**	**2**	4	2	5	2	12	1	8	7	6	4	7	1	5	0	3	0	4	1	**75.00**%	0.874
Tangent Distance (TD)	6	1	5	5	7	5	7	0	12	1	7	1	4	1	3	3	3	2	4	2	72.22%	**0.876**

N_u - number of groups with unique shape; N_o - number of groups without outliers; N_d - number of dominant shape designs; N_e - number of outliers; Ac - accuracy; R_i - Rand index.

clusters. This can result from the fact that, when there is a slight discontinuity of the exterior wall, the measured distance from the points in the perimeter dilutes such difference. This is a benefit in shapes requiring denoising; however, in datasets with no noise the results are not so good.

In the case of the Turning Function (TF) descriptor, other problem occurs. Namely, due to the absence of information in the descriptor vector resulting from wall recesses smaller than the distance between feature points—when the wall turns a small distance and turns back to the same direction. In this situation, and because this descriptor only captures the angle of the wall, the information before and after the wall change is the same. The only way to include that information is to have a feature point of the shape silhouette in it. Additionally, even if the wall recess is somehow captured, it only represents a few values in the vector as the main parts of the wall continue to have the same angle. This would be avoided only if the descriptor also measured the wall distance to a reference point.

In the results for the GB descriptor the problem is different. In this case, the descriptor vector is very sensitive to the measuring points in the grid. Therefore, if there are small variants in the shape proportions then a row of points can turn from 0 to 1 and vice-versa. For instance, a wider rectangle, when scaled to fit the measuring grid, will result in a smaller height thus having less area filled in the grid. Despite the shape being basically the same, this will result in different vectors (compare the FPD 23 in group **A** and group **F** in Fig. 8 as an example of this issue). However, when dealing with adjusted aspect ratio, the performance improves for the GB descriptor.

The TD descriptor presents the best results for both variants of the aspect ratio. This is due to the fact that it incorporates the advantages of the PD and TF descriptors, namely the ability to capture the distance of the segment and the angle change of the walls, respectively. However, when extending the use to shapes such as the equilateral triangle, square, pentagon, or other regular polygons (even a circumference), the TD descriptor will classify all of them in the same group, as the polygon tangents all have the same distance to the centre. Another issue was found with this descriptor. In some cases, when designs have the same shape type, it may consider distinct due to the sensitivity over the size of the segments in every turn of the exterior wall (see groups **F** and **G** in Fig. 9 as an example).

In the case of the distance-based descriptors (PD and TD), it is possible to control their sensitivity to wall recesses in the shape perimeter by exponentiating the normalised distances. If the exponent is lower than 1, the representation reduces the sensitivity to small variations; otherwise, when greater than 1, this is increased.

It is interesting to observe that the descriptors that have the best results are all perimeter-based representations. Area-based representations, such as the GB descriptor, are too sensitive to small changes in the proportions of the shape. This approach may have better results in shapes that require denoising. However, in synthetic datasets such as the one illustrated in the showcase, area-based representation is a less reliable approach. Limitations of these descriptors may be summarised as follows:

- PD, TF, and GB descriptors are insensitive to small recesses in the perimeter;
- TF descriptor may not capture perimeter turns if the shape's silhouette step is bigger that the turn segment dimension;
- GB descriptor greatly depends on the grid resolution thus making it very sensitive to small variations in the shape proportions;
- TD descriptor may suffer from excessive sensitivity to the segments size in wall turns, thus leading to cluster designs in different groups despite having the same shape type;
- TD descriptor clusters regular polygons (triangle, square, circle, etc.) as the same shape; and,
- TD descriptor is very sensitive to shapes with noise in the perimeter.

The matching and clustering of floor plan designs has some possible applications. One of those is to use it as a clustering mechanism for results obtained from generative design methods—for example, the EPSAP algorithm already includes these mechanisms to organise data to be presented to the decision-maker. Another example is to use it within the evolving process of population-based methods. This may have two purposes. First, to select the best individuals of each group to be kept in the next generation, thus preserving the population diversity and avoiding the dominance of one shape type. Secondly, to conduct the generative process on solutions that are of interest to the user according to their defined shape type criterion. Nowadays floor plan generative methods deal with building boundaries as defined polygons. However, if the user is able to choose the aimed shape or shapes, the method may focus only on that range of candidate designs, thus reducing the computation burden by avoiding the production and evaluation of irrelevant solutions. Finally, a possible application is to use it as a retrieval process of designs in architectural design databases.

5. Conclusion

Four shape descriptors were used to capture the form of a synthetic dataset of floor plan designs and a comparison of their performance was carried out. Every descriptor had the same vector length and the same clustering algorithm was used to aggregate the floor plans.

The perceptual analysis carried out on the four descriptors shows that Tangent Distance (TD) captures better floor plan shapes and presents fewer outliers. This was due to the fact that this

表6 描述符性能。

			Group																			
			A		**B**		**C**		**D**		**E**		**F**		**G**		**H**		**I**			
Descriptor	N_u	N_o	N_d	N_e	N_d	N_e	N_d	N_e	N_d	N_e	N_d	N_e	N_d	N_e	N_d	N_e	N_d	N_e	N_d	N_e	Ac	R_i
(a) Fixed aspect ratio																						
Point Distance (PD)	**6**	0	5	4	8	6	6	2	9	1	7	2	5	3	5	1	3	1	3	1	70.83%	0.861
Turning Function (TF)	**6**	**2**	3	2	4	3	8	7	8	0	6	4	7	6	4	1	5	0	3	1	66.67%	0.842
Grid-Based (GB)	5	1	3	7	4	4	8	2	5	5	3	2	4	0	8	4	2	2	3	6	55.56%	0.824
Tangent Distance (TD)	**6**	1	5	4	8	6	7	0	10	1	7	1	3	2	5	1	4	2	4	2	**73.61**%	**0.873**
(b) Non-fixed aspect ratio																						
Point Distance (PD)	6	1	5	4	5	4	7	0	10	1	7	6	4	2	5	3	3	3	2	1	66.67%	0.852
Turning Function (TF)	6	1	5	3	4	0	5	4	9	1	5	4	7	4	3	1	8	1	4	4	69.44%	0.858
Grid-Based (GB)	**7**	**2**	4	2	5	2	12	1	8	7	6	4	7	1	5	0	3	0	4	1	**75.00**%	0.874
Tangent Distance (TD)	6	1	5	5	7	5	7	0	12	1	7	1	4	1	3	3	3	2	4	2	72.22%	**0.876**

N_u - 具有独特形状的簇数；N_o - 无异常值的簇数；N_d - 占主导地位的形状模式数；N_e - 异常值数；Ac - 准确性；R_i - Rand指数。聚类。这可能是由于当外墙存在轻微不连续时，周界点测得的距离会平滑掉这种差异所致。对于需要去除噪声的形状，能够很好地捕捉到这种差异。然而，在不含噪声的数据集中所使用的合成数据集这类场景下，基于区域的表示方法则显得不够可靠。这些描述符的局限性可归纳如下：

结果并不理想。

就转向函数（TF）描述子而言，还会出现另一类问题。具体而言，当墙体的凹陷小于特征点之间的距离时，描述子向量中会缺失相关信息——即墙体仅发生小幅转折后又恢复至原方向的情况。在这种情形下，由于该描述子仅编码墙体的朝向角，墙体转折前后所携带的信息完全相同。要纳入这部分信息，唯一的办法是在形状轮廓上设置一个特征点。此外，即便墙体的凹陷被某种方式捕捉到，它在向量中也仅表现为少数几个值，因为墙体的主体部分仍保持相同的朝向角。只有当描述子同时度量墙体到参考点的距离时，这一问题才能得以避免。

在GB描述符的实验结果中，问题则有所不同。在这种情况下，描述符向量对网格中的采样点极为敏感。因此，即使形状比例发生微小变化，一整行的采样点也可能由0变为1，反之亦然。例如，一个较宽的矩形在缩放以适配测量网格时，其高度会相应减小，从而导致网格中被填充的面积减少。尽管形状本质上并无差异，但这仍会导致生成不同的描述符向量（如图8中A组与F组的FPD 23所示，即为此类问题的典型示例）。然而，在采用调整后的宽高比进行处理时，GB描述符的性能有所提升。

TD描述子在两种宽高比变体中均表现出最佳性能。这是因为它同时融合了PD和TF描述子的优势，即分别能够刻画线段的长度以及墙体的角度变化。然而，当将其应用于等边三角形、正方形、正五边形或其他正多边形（甚至圆）等形状时，由于这些图形的各条切线到中心的距离均相等，TD描述子会将它们归为同一类别。此外，该描述子还存在另一个问题：在某些情况下，当设计具有相同的形状类型时，由于对外墙每一段线段尺寸的敏感性，它可能会将其判定为不同类别（参见图9中的F组和G组作为示例）。

对于基于距离的描述符（PD 和 TD），可以通过对归一化距离进行指数运算来调控其对形状周界中壁面凹陷的敏感性。当指数小于 1 时，该表示会降低对微小变化的敏感度；反之，当指数大于 1 时，则会增强这种敏感度。

有趣的是，实验结果最佳的描述子均为基于周长的表示方法。而基于面积的表示方法，如GB描述子，则对形状比例的微小变化过于敏感。这种方法可能具有

- PD、TF 和 GB 描述符对周界上的小凹陷不敏感；
- 如果形状轮廓的步长大于转弯段的尺寸，TF描述符可能无法捕捉到周界转弯。
- GB描述符在很大程度上依赖于网格分辨率，因此对形状比例的微小变化极为敏感；
- TD描述符在壁面转折处可能对分段尺寸过于敏感，从而导致即使具有相同形状类型的不同组别也被划分为不同的聚类；
- TD描述符将正多边形（如三角形、正方形、圆形等）归为同一形状；并且，
- TD描述子对边界存在噪声的形状非常敏感。

平面布局设计的匹配与聚类具有若干潜在应用。其一，可将其用作生成式设计方法所得结果的聚类机制——例如，EPSAP算法已内置此类机制，用于组织待呈递给决策者的数据。另一应用是在基于种群的方法的演化过程中加以利用，这可能具有两个目的：其一，从每一聚类中筛选出最优个体以保留至下一代，从而维持种群多样性并避免某一形态类型占据主导；其二，针对用户根据其设定的形态类型标准所感兴趣的方案开展生成过程。目前，平面布局的生成方法通常以多边形形式定义建筑边界；然而，若用户能够指定目标形态或形态类别，则该方法可仅聚焦于相应范围内的候选设计方案，从而通过避免生成与评估无关方案来降低计算负担。最后，还有一种可能的应用是将其作为建筑设计数据库中的方案检索手段。

5. 结论

采用了四种形状描述子来表征一组合成的平面布局设计数据集，并对其性能进行了比较。每种描述子的向量长度均相同，且使用了相同的聚类算法对这些平面布局进行聚合。

对四个描述子进行的感知分析表明，切线距离（TD）能够更准确地刻画平面布局形状，并且产生的异常值更少。这是由于该

descriptor not only measures the distance to the geometric centre but also captures the discontinuities in the walls. The outliers resulted from excessive sensitivity to small wall recesses in the perimeter thus shifting the design to other group with a similar overall configuration.

In the case of the other descriptors, the opposite happens. The Grid-Based (GB) descriptor presents the least reliable approach and is very sensitive to different proportions in the same shape thus designs are distributed over several groups with different dominant shapes.

For the fixed aspect ratio variant, the performance of the two best descriptors was a Rand index of 0.861 and 0.873 for the Point Distance (PD) and TD, respectively. In the non-fixed aspect ratio descriptor variant, the descriptors with the best performance were the GB and TD, with a Rand index of 0.874 and 0.876, respectively.

Despite these good results, some issues still need to be tackled. Future work includes extending these approaches to non-orthogonal and multi-storey designs, to study other descriptors that capture the inner space relations in the floor plan, and to test the performance of descriptors in other types of clustering algorithms.

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Appendix A. Descriptors' results for non-fixed aspect ratio

Figs. A.10, A.11, A.12, and A.13 display the resulting clustering of each of the four shape representations with non-fixed aspect ratio.

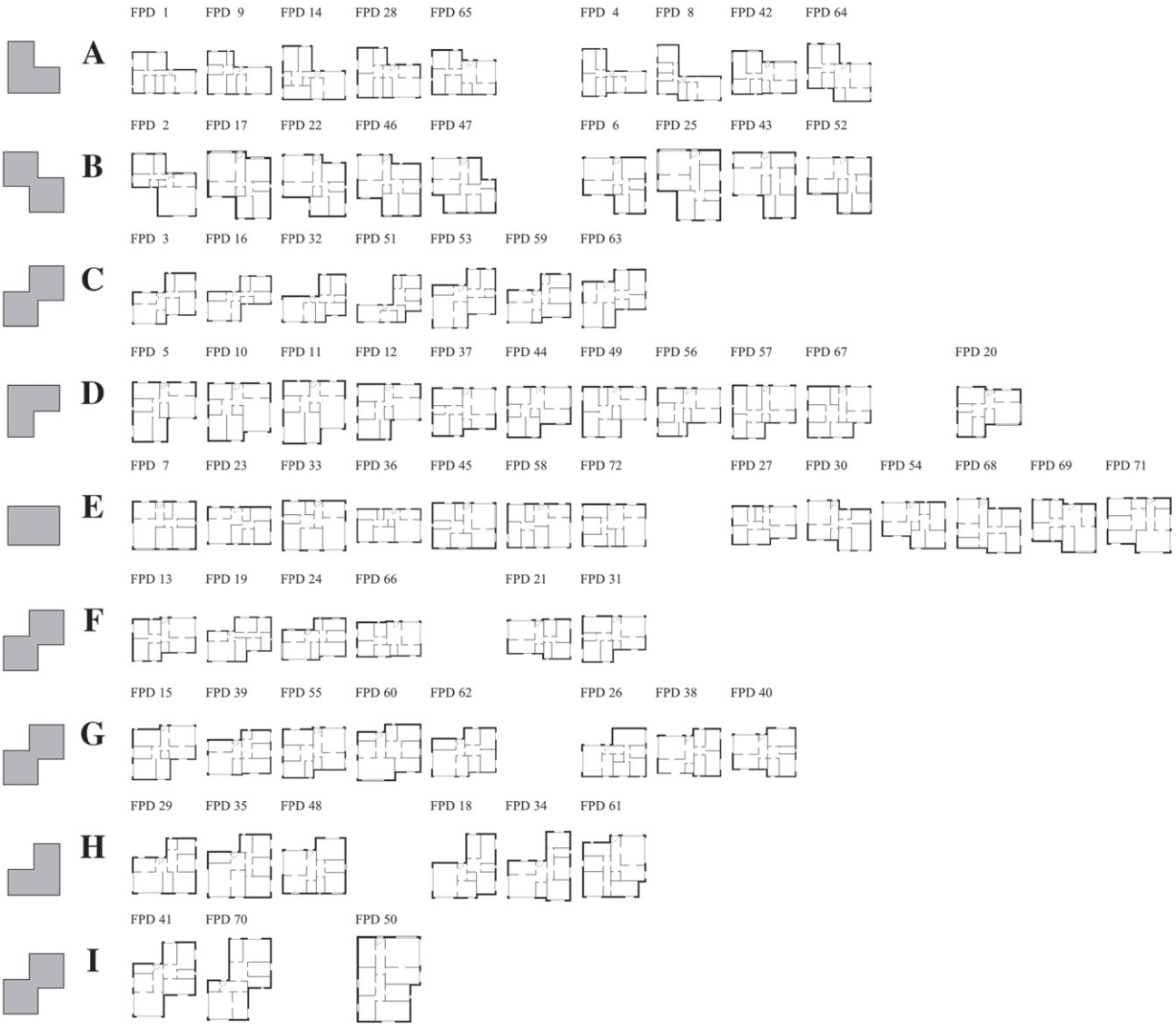


Fig. A.10. Clustering results using Point Distance (PD) descriptor with non-fixed aspect ratio.

该描述符不仅度量与几何中心的距离，还能够捕捉墙体的不连续性。由于对周界上微小的墙体凹陷过于敏感，导致部分样本被误判为异常值，从而将其归入具有相似整体构型的另一类别。

对于其他描述符而言，情况则相反。基于网格的（GB）描述符是可靠性最低的方法，并且对同一形状中不同比例的变化极为敏感，因此设计会被分配到多个具有不同主导形状的簇中。

对于固定宽高比变体，两种最优描述符的性能分别为：点距离（PD）的Rand指数为0.861，TD的Rand指数为0.873。在非固定宽高比描述符变体中，性能最优的描述符是GB和TD，其Rand指数分别为0.874和0.876。

尽管取得了这些良好结果，仍有一些问题有待解决。未来的研究工作将包括将这些方法扩展至非正交情形。

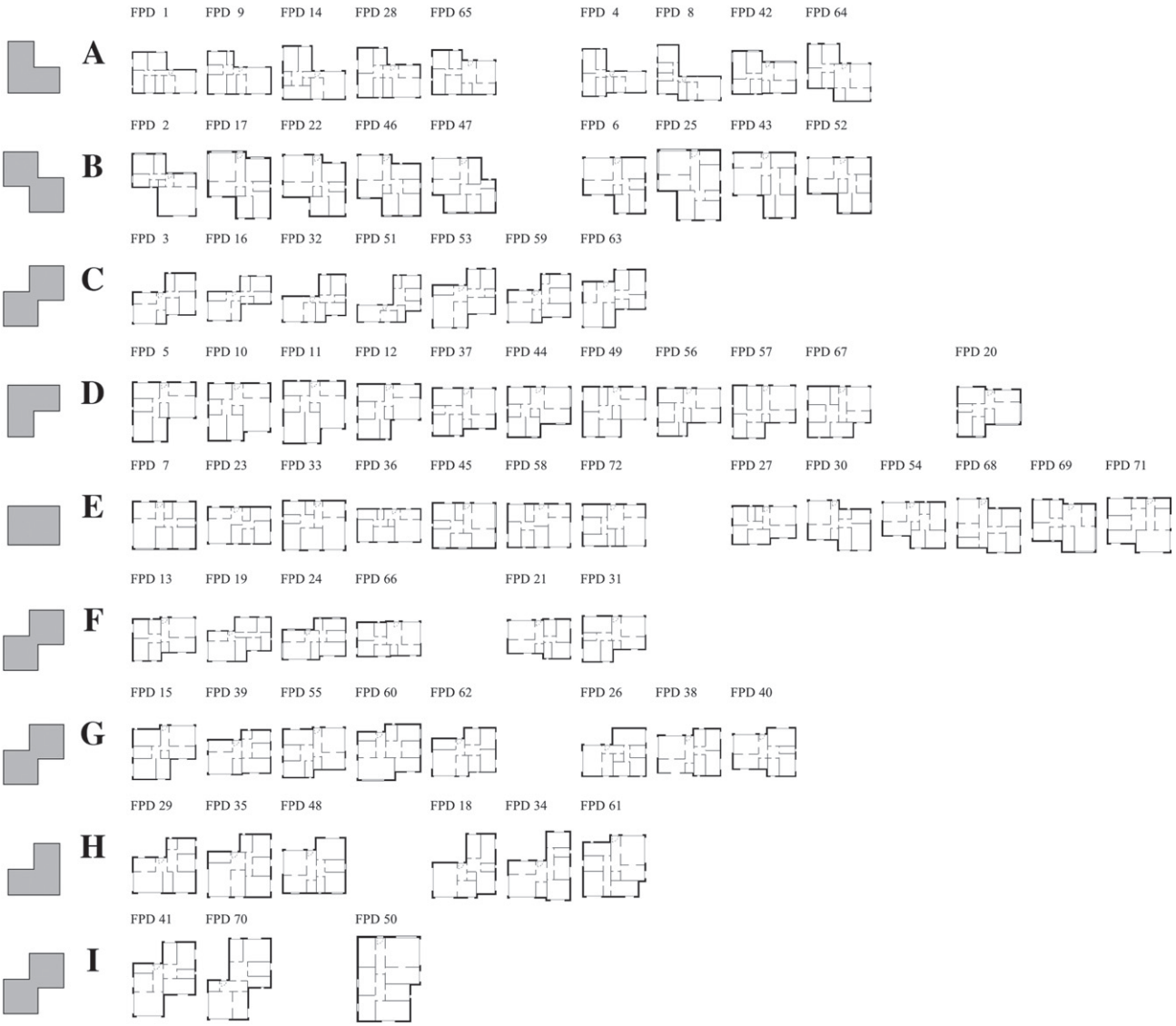
以及多层建筑布局，以研究能够刻画平面图中内部空间关系的其他描述子，并在其他类型的聚类算法中评估这些描述子的性能。

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附录A. 非固定宽高比的描述符结果

图A.10、A.11、A.12和A.13展示了在非固定长宽比条件下，四种形状表示方法所得到的聚类结果。



图A.10 使用点距离（PD）描述子且宽高比不固定的聚类结果。

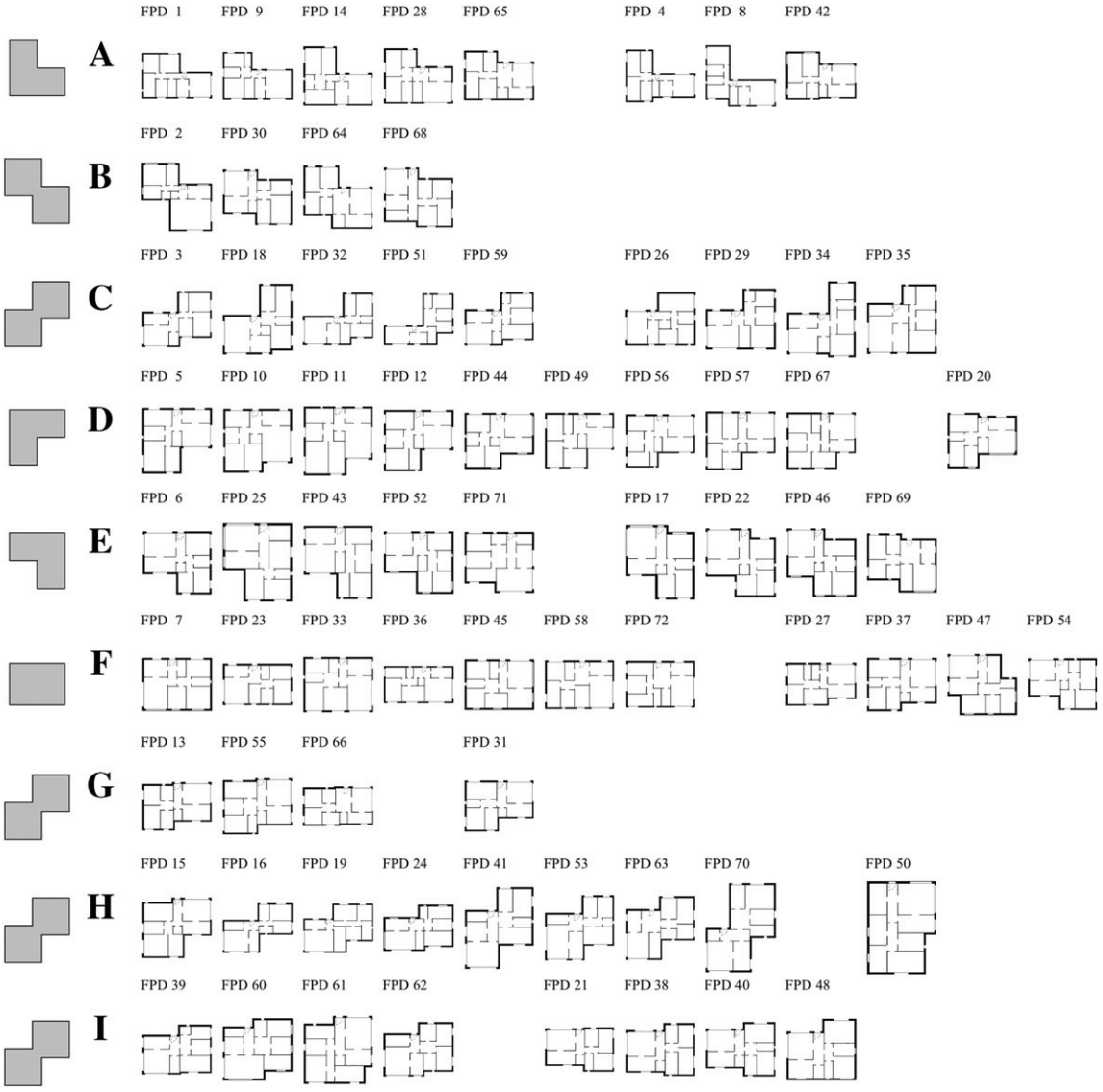
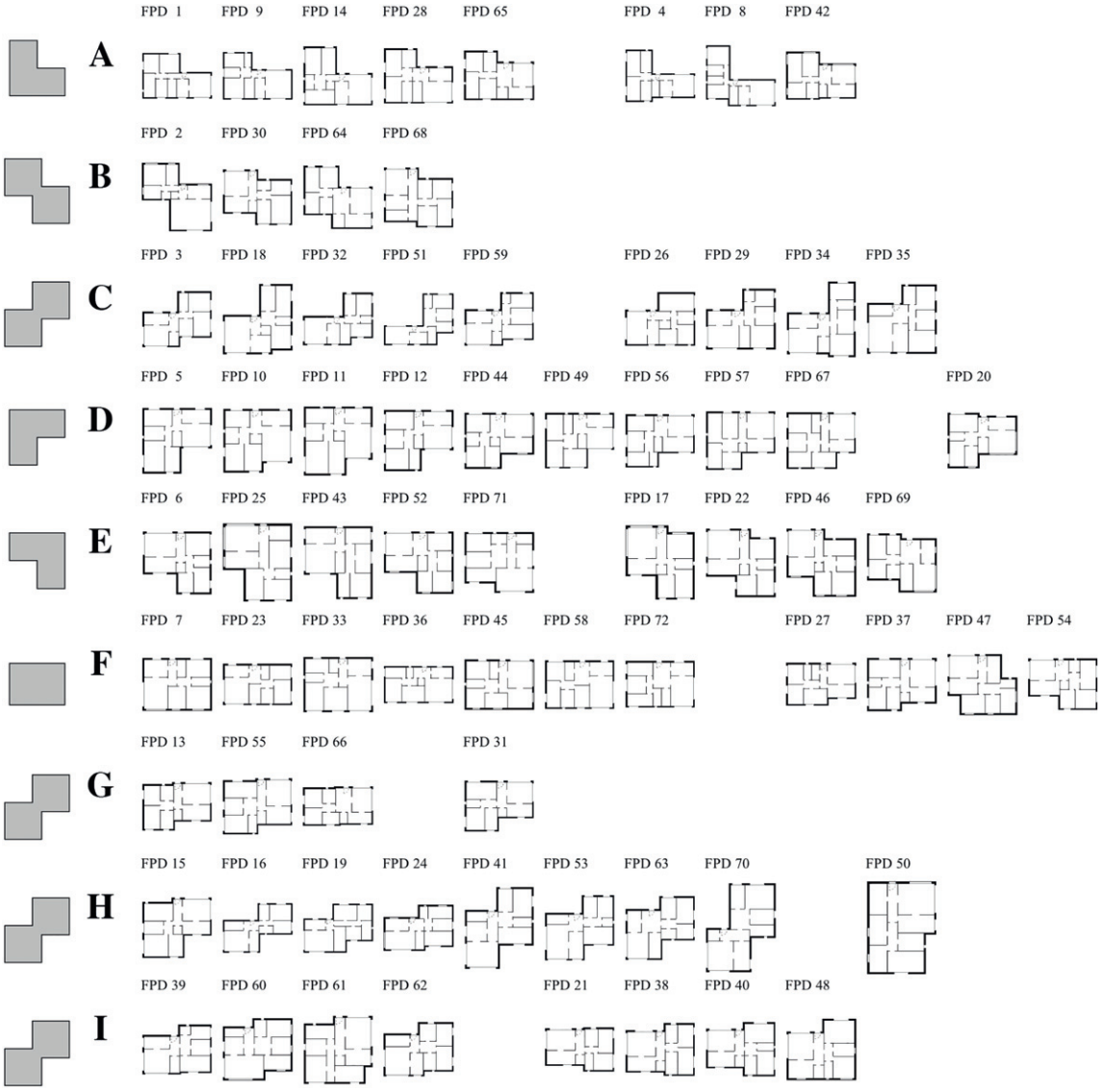


Fig. A.11. Clustering results using Turning Function (TF) descriptor with non-fixed aspect ratio.



图A.11. 使用非固定型转向函数（TF）描述符的聚类结果 比例。



Fig. A.12. Clustering results using Grid-Based (GB) descriptor with non-fixed aspect ratio.



图A.12 使用非固定宽高比的基于网格（GB）描述符得到的聚类结果。



Fig. A.13. Clustering results using Tangent Distance (TD) descriptor with non-fixed aspect ratio.

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图A.13：采用切线距离（TD）描述子且宽高比不固定时的聚类结果。

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